Swiftradish Phase 2: Reconciled Product Experience, System Architecture, and Implementation Specification

A. Input Inventory and Phase 1/Blueprint Extraction

1.1 Input Inventory and Source-of-Truth Coverage Map

This Phase 2 specification reconciles the supplied research prompt and project handoff constraints against the 73-page prior blueprint and current primary vendor and statutory documentation. In accordance with the project truth hierarchy, explicit user constraints and verified primary documentation supersede prior blueprint recommendations wherever conflicts arise.

Artifact / Input Source	Status in Workspace	Material Established ([USER DECISION] / [VERIFIED])	Material Proposed Only ([RECOMMENDATION])	Gaps, Duplications, or Unverified Assumptions Identified
SWIFTRADISH_PHASE_2_GEMINI_RESEARCH_PROMPT.md (consolidating SWIFTRADISH_CONTEXT.md, Phase 2, and Phase 3 requirements)	**Present & Complete**	Project is a clean-slate restart; solo builder using Cursor Pro (Grok and Composer 2.5 available); Python/FastAPI preferred; Supabase project, Backblaze B2 bucket, and GitHub repo (Swiftradish) created; local repo not yet linked to origin.	Staged vertical-slice execution; Stitch used strictly for post-decision UI exploration rather than product truth.	Actual cloud configurations, billing tiers, IAM/bucket policies, and region settings for the user's Supabase and Backblaze B2 accounts remain uninspected ([UNVERIFIED]).
Swiftradish Product and Engineering Blueprint.pdf (Prior 73-page Gemini Argon output)	**Present & Complete**	Competitor taxonomy (Instagram, Discord, Reddit, Pinterest, Strava, BeReal, Geneva); Next.js serverless WebSocket limitations;	Co-active thesis ("Co-Plots" + "Radish Patches"); 40-screen catalog; "Nocturnal Botanical Studio" dark theme; 216-pose "Pip the Radish" taxonomy;	**Major Conflict:** Unconditionally replaces the user's Backblaze B2 bucket with Cloudflare R2 for media storage. **Scope Overload:** Mandates 40

Artifact / Input Source	Status in Workspace	Material Established ([USER DECISION] / [VERIFIED])	Material Proposed Only ([RECOMMENDATION])	Gaps, Duplications, or Unverified Assumptions Identified
		PostgreSQL tsvector + pg_trgm hybrid indexing capabilities; iOS 16.4+ PWA standalone Web Push rules.	14-table UUIDv7 schema; 12-phase roadmap.	screens, 3 Co-Plot modes, full DM/group chat, 30 mascot poses, and badge engines in the initial solo-builder MVP.
Primary External Vendor & Legal Documentation (2025–2026)	**Verified via Primary Sources**	Backblaze B2 pricing ($0.006/GB/mo), S3 presigned PUT support, presigned POST limitation, and $0 egress via Cloudflare Bandwidth Alliance; Supabase asymmetric JWKS (ES256) at /auth/v1/.well-known/jwks.json; India DPDP Rules notified Nov 13–14, 2025 with phased enforcement through May 14, 2027; India IT Rules 2021 Rule 3 timelines.	Specific Cloudflare Worker header-signing patterns for private B2 buckets; PyJWT[crypto] algorithm pinning to mitigate CVE-2026-102266.	Specific Indian legal determinations (SSMI thresholds, Eighth Schedule language translations, and parental verification exemptions) require review by qualified Indian legal counsel.

1.2 Baseline Extraction from the Prior Blueprint

The prior blueprint proposes positioning Swiftradish between passive algorithmic broadcast networks (Instagram, TikTok) and ephemeral, unsearchable chat platforms (Discord, WhatsApp) by introducing **Co-Plots** (posts upgraded with interactive coordination modules) and **Radish Patches** (topic communities constrained to a visual feed, an event calendar, and up to three chat channels). It defines two home feeds—**My Patch** (strictly chronological) and **Fresh Harvest** (a time-decayed relevance feed with provenance badges)—alongside a 40-screen catalog (SCR-01 through SCR-40), a dark-first token palette (#0B0D11 canvas, #FF4D5E primary accent, #2EE59D secondary accent), a 14-entity PostgreSQL 16 schema using UUIDv7 primary keys, a Python 3.12+ / FastAPI 0.137+ modular monolith with arq background workers,

and a 12-phase roadmap. While its domain modeling and state-machine rigor are strong, several infrastructure, scope, and regulatory assertions conflict with the solo builder's constraints and verified 2026 primary documentation, requiring systematic reconciliation.

B. Reconciliation Ledger

Every material conflict, unverified assumption, or architectural divergence between the prior blueprint, the project context, and primary sources is resolved in the ledger below.

ID	Blueprint Claim or Proposed Decision	Verified Evidence & Primary Source Reality	Identified Conflict or Gap	Reconciled Specification Treatment & Label	Confidence
REC-01	**MVP Scope:** Ship all 40 screens, 3 Co-Plot modes, full 1:1/group WebSocket DMs, 3-channel Patch chat, Web Push, and 30 Pip SVG/Lottie poses before public launch.	User is a solo builder using Cursor Pro with Grok and Composer 2.5 ([USER DECISION]). Large, coupled surface areas cause context degradation in non-frontier coding models.	Building 5 complex subsystems simultaneously creates unacceptable schedule and reliability risk for a solo developer.	**[RECOMMENDATION] (Revised MVP Boundary):** Partition the 40 screens into **MVP Launch Core (22 screens, Slices 1–6)** and **Stage 2 Fast-Follow (18 screens, Slices 7–9)**. Unify Co-Plots into one cohesive card (Activity RSVP + Shared Carousel + single Plot Thread). Defer standalone 1:1 DM inbox (SCR-27/29), 3-channel Patch lounges, and badge engines to Stage 2.	**High**
REC-02	**Object Storage:** Use Cloudflare R2 ($0.015/GB/mo) for all active	User already created a Backblaze B2 bucket ([USER DECISION]).	Blueprint ignores the user's provisioned B2 bucket and	**[RECOMMENDATION] (Retain Backblaze B2 + Cloudflare**	**High**

ID	Blueprint Claim or Proposed Decision	Verified Evidence & Primary Source Reality	Identified Conflict or Gap	Reconciled Specification Treatment & Label	Confidence
	media and relegate Backblaze B2 solely to nightly database backups (ADR-06).	B2 storage costs $0.006/GB/mo (60% cheaper than R2), includes 3x free egress, and provides **100% free ($0.00) egress via Cloudflare CDN/Workers under the Bandwidth Alliance** ([VERIFIED]).	overpays 2.5x per GB of storage on R2 without justification.	**CDN):** Use the existing Backblaze B2 bucket as primary object storage behind a Cloudflare Bandwidth Alliance CDN/Worker proxy, retaining R2 as a zero-code-change S3 fallback via aioboto3 configuration.	
REC-03	**Direct Upload Protocol:** Browser uploads directly to storage via S3-compatible presigned URLs.	Backblaze B2's S3-compatible API supports presigned PUT URLs (PutObject) and AbortMultipartUpload, but **does not support browser uploads via presigned POST policies** ([VERIFIED]).	Any implementation attempting generate_presigned_post against B2 will fail at runtime.	**[VERIFIED] Contract:** Mandate generate_presigned_url('put_object', ...) with strict Content-Type and Content-Length headers across all client upload flows.	**High**
REC-04	**Private Media CDN Delivery:** Serve public media via CDN and private media via 15-minute signed URLs.	Cloudflare does not cache responses when the origin request includes an Authorization header unless configured via a Cloudflare Worker (backblaze-b2-samples/cloudfl	Direct B2 URLs expose bucket names and consume direct egress quota if not routed through Cloudflare ([VERIFIED]).	**[RECOMMENDATION]:** Route public variants (thumb_400w, feed_1080w, full_2048w) through a Cloudflare Worker bound to cdn.swiftradish.app that signs origin requests	**High**

ID	Blueprint Claim or Proposed Decision	Verified Evidence & Primary Source Reality	Identified Conflict or Gap	Reconciled Specification Treatment & Label	Confidence
		are-b2) and Bucket Info {"Cache-Control":"public"} ([VERIFIED]).		to a private B2 bucket, preventing direct B2 URL enumeration while preserving edge caching and $0 egress.	
REC-05	**Supabase Scope & JWT Auth:** Verify Supabase JWTs locally in FastAPI via JWKS/HS256 (SR-06, ADR-03).	Supabase exposes asymmetric signing keys (ES256) at https://<project-id>.supabase.co/auth/v1/.well-known/jwks.json ([VERIFIED]). Allowing both HS256 and ES256 enables public-key HMAC forgery (CVE-2026-102266), and deriving the JWKS URL from an unverified token iss claim enables SSRF/spoofing ([VERIFIED]).	Blueprint mentions JWKS/HS256 interchangeably without algorithm pinning or iss trust-anchor rules.	**[VERIFIED] Security Mandate:** Use the existing Supabase project for PostgreSQL 16 + Supabase Auth. In FastAPI, install pyjwt[crypto], hardcode SUPABASE_JWKS_URL and EXPECTED_ISSUER in server configuration, and restrict jwt.decode() strictly to algorithms=["ES256"] and audience="authenticated".	**High**
REC-06	**India DPDP Act 2023 Status:** Enforces 18+ age gate under Section 9 of the DPDP Act 2023.	MeitY officially notified the **DPDP Rules, 2025** on **13–14 November 2025** (G.S.R. 843(E), 844(E), 846(E)), establishing the	Blueprint omits the notified DPDP Rules 2025, Rule 7 mandatory 72-hour breach reporting (zero materiality threshold),	**[VERIFIED] Legal Alignment:** Retain the 18+ age gate as a technical risk control, add standalone itemized	**High**

ID	Blueprint Claim or Proposed Decision	Verified Evidence & Primary Source Reality	Identified Conflict or Gap	Reconciled Specification Treatment & Label	Confidence
		Data Protection Board immediately, Consent Managers by **14 Nov 2026**, and full substantive enforcement by **13–14 May 2027** (with a Jan 2026 consultation proposing acceleration to Nov 2026) ([VERIFIED]).	absence of "legitimate interest", and 22-language notice rules.	consent notices, implement a 72-hour breach-notification admin runbook, and flag DigiLocker age verification and Eighth Schedule translations for Indian legal counsel.	
REC-07	**India IT Rules 2021 SLAs:** 24h acknowledgment, 24h severe takedown, 72h law enforcement SLA, 15d general resolution, and Rule 4(8) dispute notice.	Under **Rule 3(2)(a)(i)** (as amended Oct 28, 2022 via G.S.R. 794(E)), user complaints requesting removal of prohibited content under **Rule 3(1)(b)** (except sub-clauses (i), (iv), (xi)) must be resolved within **72 hours**, not just law enforcement orders ([VERIFIED]). Rule 4(8) applies statutorily to SSMIs (>50 lakh users).	Blueprint mislabels the 72-hour SLA as primarily a law enforcement timeline rather than a mandatory user-report takedown window for Rule 3(1)(b) harms.	**[VERIFIED] SLA Correction:** Map all Rule 3(1)(b)(ii)–(x) user content-removal reports to STANDARD_72H, Rule 3(2)(b) bodily privacy/non-consensual imagery/impersonation reports to CRITICAL_24H, law enforcement orders (Rule 3(1)(j)) to LEGAL_72H, and IP/ownership/general grievances to	**High**

ID	Blueprint Claim or Proposed Decision	Verified Evidence & Primary Source Reality	Identified Conflict or Gap	Reconciled Specification Treatment & Label	Confidence
				GENERAL_15D, with 180-day tamper-evident preservation (Rule 3(1)(g)).	
REC-08	**Terminology & Onboarding Cognitive Load:** Introduces "Co-Plots", "Radish Patches", "My Patch", "Fresh Harvest", "Sprouts", and "Pip" simultaneously.	Novelty metaphors increase cognitive friction if users cannot grasp the core action without a tutorial ([INFERENCE]).	Four overlapping botanical metaphors on the home screen obscure standard social actions.	**[RECOMMENDATION] (Terminology Simplification):** Pair every botanical brand term with an explicit functional subtitle in the UI: **Patches (Communities)**, **Co-Plots (Shared Activities & Albums)**, **Following Feed (My Patch)**, and **Discover Feed (Fresh Harvest)**.	**High**

C. Product Thesis and MVP Boundary

3.1 Reconciled Product Thesis and Product Narrative

The revised central product thesis ([RECOMMENDATION] — Revised for Clarity) establishes that Swiftradish is a co-active social platform where interest-driven communities ("Patches") and mutual friends turn rich multi-image posts directly into shared activities, collaborative photo carousels, and contextual group coordination ("Co-Plots") within a single web and PWA surface. Existing platforms force groups to split their social lives across passive algorithmic feeds (Instagram, Pinterest) where ideas are discovered and ephemeral chat threads (WhatsApp, Discord) where plans and post-event photos get buried in scrollback. Swiftradish eliminates this context switch by making the post itself the coordination hub: any user publishing a post or multi-image carousel can toggle it into a **Co-Plot**, attaching a structured RSVP roster, a contextual discussion thread, and a shared carousel where approved participants append their own photos during or after the activity.

3.2 Target User Segments, Jobs-to-be-Done, and Switching Triggers

User Segment	Core Job-to-be-Done (JTBD)	Trigger Situation	Current Fragile Workaround	Reason to Switch to Swiftradish
1. Urban Social & Campus Coordinators (Primary MVP Target, Ages 18–28)	Organize a weekend hike, café hop, photo walk, or study jam and collect everyone's photos afterward in one permanent place.	Planning a group outing for 4–25 friends or club members and wanting both a clear headcount and a shared post-event photo dump.	Screenshot idea on Instagram -> paste into WhatsApp group -> lose RSVP count in 80 chat messages -> beg attendees for Google Drive links days later.	One **Co-Plot card** handles the visual invite, 1-tap RSVP (Going/Maybe), contextual thread, and post-event **Shared Memory Carousel**.
2. Niche Hobbyist & Creative Collectives (Primary MVP Target, Ages 18–34)	Run a local or thematic club (e.g., analog photography, urban sketching, indie game dev, run club) with discoverable visual showcases.	Weekly creative prompt or local chapter meetup where members want to showcase work without it vanishing in Discord scrollback.	Maintain a fragmented Discord server (poor visual browsing), an Instagram page (suppressed reach), and a Notion/Luma calendar.	A **Radish Patch (Community)** provides a clean visual feed, an upcoming Co-Plots tab, and deterministic chronological delivery to members.
3. Everyday Visual Curators (Secondary Segment)	Share multi-image photo essays and curated notes with followers without algorithmic video penalties.	Returning from a trip or completing a project and wanting to publish a 10-slide carousel with formatted text notes.	Post to Instagram (buried unless formatted as a Reel) or Reddit (text-heavy, anonymous hostility).	Deterministic chronological delivery in **Following (My Patch)** plus transparent, topic-seeded discovery in **Discover (Fresh Harvest)**.

3.3 Core User Loop and Minimum Social/Content Graph

The canonical Swiftradish user loop operates across four deterministic stages ([RECOMMENDATION], High Confidence). In the initial **Trigger & Discovery** phase, a user enters via a shared Co-Plot deep link (/p/[postId]) or browses a topic **Patch** (/patches/[slug]) seeded during onboarding. During the **Co-Active Commitment** phase, instead of registering a passive "like," the user taps **Join Co-Plot** (Going) on an upcoming activity or visual prompt, which immediately grants access to the Co-Plot's contextual coordination thread. In the **Collaborative Contribution** phase, during or after the activity, participants upload their own photos directly into the parent Co-Plot's **Shared Memory Carousel**, which are auto-approved for mutuals or approved via a 1-tap review by the host. Finally, in the **Meaningful Return** phase, every contributor is credited on their specific carousel slide (@handle), the updated Co-Plot

surfaces in participants' feeds and profile archives, and high-signal notifications bring members back when new slides or follow-up Co-Plots are planted.
Supporting this core loop requires a strictly bounded minimum social and content graph ([RECOMMENDATION]). The **User Node** (users, user_profiles) establishes a public or private identity with a unique handle, bio, and avatar. The **Hybrid Relationship Edge** (user_relationships) manages asymmetric FOLLOW edges for public discovery alongside automatically computed reciprocal is_mutual = TRUE edges that unlock frictionless Co-Plot slide contributions. The **Community Container** (patches, patch_members) provides a topic space anchoring public and member posts and Co-Plots so new users encounter immediate activity even before building a personal friend graph. Finally, the **Actionable Artifact** (posts, co_plots, co_plot_participants, media_assets) unifies standard or Co-Plot posts supporting 1 to 10 images, structured RSVP state, contributor slide submissions, reactions, and 3-level nested comments and discussion.

3.4 Scope Challenge and Reconciled Feature Decision Matrix

Evaluating the blueprint's feature list against solo-builder execution reality with Cursor Pro (Grok / Composer 2.5) requires ruthlessly pruning secondary surfaces from the initial launch while keeping the core Co-Plot loop 100% intact ([RECOMMENDATION], High Confidence).

Feature ID & Name	Blueprint Tier	Reconciled Phase 2 Decision	User-Value Consequence & Engineering Rationale for Solo Builder
F-01: 18+ Auth, Sessions (Supabase Auth + Local FastAPI JWKS) & Profiles	Tier 1	**Keep in MVP Core (Slice 1)**	Foundational security, identity, and DPDP Section 9 age-gating prerequisite.
F-02: Hybrid Social Graph (Asymmetric Follow + Auto-Mutual Detection + Private Accounts)	Tier 1	**Keep in MVP Core (Slice 2)**	Essential for distinguishing public followers from trusted mutual collaborators.
F-03: Block, Mute, Report & Statutory Grievance Flow (IT Rules 2021 SLAs)	Tier 1	**Keep in MVP Core (Slice 6)**	Legally mandatory for Indian launch under IT Rules 2021 Rule 3 and essential against harassment.
F-04: Rich Multi-Image Carousel Composer (Up to 10 Images, Direct-to-B2 Presigned PUT, pyvips EXIF Strip)	Tier 2	**Keep in MVP Core (Slice 3)**	Primary creative expression primitive; direct-to-B2 upload keeps FastAPI stateless and memory-safe.
F-05: Collaborative "Co-Plot" Cards (Activity RSVP + Shared Memory	Tier 2	**Keep in MVP Core — Simplified (Slice 4)**	**Core thesis differentiator.** Combine Activity RSVP and Shared Carousel

Feature ID & Name	Blueprint Tier	Reconciled Phase 2 Decision	User-Value Consequence & Engineering Rationale for Solo Builder
Carousel)			into one unified Co-Plot model; defer the separate "Visual Remix Strip" sub-mode to Stage 2.
F-06: Dual-Mode Home Feed (Following / My Patch Chronological + Discover / Fresh Harvest Ranked)	Tier 2	**Keep in MVP Core (Slice 5)**	Solves cold-start discovery via deterministic SQL scoring while guaranteeing chronological reach for followed accounts.
F-07: Contextual Reactions, 3-Level Nested Comments & Bookmarks	Tier 2	**Keep in MVP Core — Simplified (Slice 5)**	Keep 6 core radish reactions, 3-level comments, and bookmarks. Defer "Quote-Remix into new Co-Plot" to Stage 2.
F-09: Radish Patches (Topic Communities) & PostgreSQL tsvector + pg_trgm Unified Search	Tier 2	**Keep in MVP Core — Simplified (Slice 5)**	Essential for cold-start content density. Constrain MVP Patches to **Curated Feed + Co-Plots Tab**; defer the "3 separate real-time chat channels per Patch" to Stage 2.
F-08: Real-Time WebSockets (Co-Plot Coordination Thread vs Standalone 1:1/Group DM Inbox)	Tier 2	**Stage Across Slice 4 & Stage 2 (Slice 7)**	In **MVP Core**, ship the **Co-Plot Discussion Thread** (backed by conversations and messages with WebSocket live updates and REST fallback). Defer standalone 1:1 DM Inbox, Message Requests Quarantine, and 15-min message edits to **Stage 2 (Slice 7)** once core Co-Plots are stable.
F-10: In-App Activity Center & PWA Web	Tier 2	**Keep In-App in MVP Core (Slice 6); Defer**	In-app deduplicated notifications (SCR-31)

Feature ID & Name	Blueprint Tier	Reconciled Phase 2 Decision	User-Value Consequence & Engineering Rationale for Solo Builder
Push Notifications		**Web Push to Stage 2 (Slice 8)**	ship in MVP Core. VAPID Web Push and iOS Home Screen Primer (SCR-10) ship immediately after in Stage 2 (Slice 8).
F-11: "Pip the Radish" Mascot & Milestone Badges	Tier 2	**Simplify in MVP Core (8 Static SVGs); Defer 30-Pose/Lottie & Badge Engine to Stage 2**	Commissioning 30 SVGs + 12 Lotties + anti-farming badge SQL triggers blocks MVP launch. Ship 8 static SVGs for empty/error/onboarding states in MVP.
F-12, F-13, F-14: Short Video (60s Cloudflare Stream), 24h Ephemeral "Sprouts", 216-Pose Library	Tier 3 / 4	**Defer Pending Evidence (Post-Launch)**	Gate behind explicit beta validation hypotheses to prevent bandwidth and transcoding burn before PMF.
F-15, F-16: Creator Subscriptions/Ticketing, Long Video (>60s), WebRTC Audio/Video Calls, E2EE	Tier 5	**Reject / Remove**	Severe KYC/GST/TURN server drag; E2EE breaks server-side search and statutory IT Rules moderation.

3.5 Cold-Start Plan, Explicit Non-Goals, and Validation Hypotheses

A co-active platform fails if a user logs into an empty feed or plants a Co-Plot with zero respondents ([RECOMMENDATION]). Cold start is mitigated through three coupled mechanics. First, through **Pre-Seeded Starter Patches**, before onboarding the first external cohort, the founder seeds ৩ताठ high-signal **Radish Patches** (#hyderabad-photo-walks, #indie-builders, #urban-sketching, #weekend-trail-runs, #specialty-coffee, #analog-film, #book-swaps, and #boardgame-nights), each populated with 3 to 5 authentic founder or ambassador posts and 1 active weekend Co-Plot, strictly utilizing real content while banning fake user counters and fabricated mock posts. Second, through **Mandatory Onboarding Patch Seeding (SCR-09)**, new registrants select at least 3 starter Patches during onboarding, ensuring Following (My Patch) and Discover (Fresh Harvest) render real community content on first load. Third, through **Single-Player-to-Multiplayer Utility via Unauthenticated Deep Links (SCR-02)**, a host can create a Co-Plot for an existing friend group and paste the canonical link (/p/[postId]) into WhatsApp or Instagram; recipients see a complete server-rendered preview with OpenGraph cards and the RSVP roster, requiring only a 30-second sign-up to tap **Join Plot** and upload their photos afterward.

To protect engineering focus, Swiftradish will explicitly avoid building an algorithmic short-video entertainment feed, daily forced camera timers (BeReal style), sprawling multi-channel Discord server hierarchies, arbitrary file-sharing lockers, or native iOS/Android binaries during the MVP phase. Furthermore, three core hypotheses require cheap pre-scale validation prior to expanding scope. Under the **Co-Plot Conversion Hypothesis**, at least 20% of weekly active users in the initial 100-user closed beta must either RSVP to a Co-Plot or submit a photo slide to a Shared Memory Carousel within 14 days of registration. Under the **Multi-Author Carousel Hypothesis**, at least 25% of hosted Co-Plots must receive photo submissions from two or more distinct participants, validating whether collaborative carousels drive retention before building 24-hour "Sprouts". Finally, under the **Video Demand Hypothesis**, the team will measure composer click attempts on a lightweight "Video (Coming Soon)" tooltip or external video link shares before provisioning paid Cloudflare Stream infrastructure.

D. Information Architecture and Screen/State Matrix

4.1 Global Navigation, Route Groups, and Responsive Behavior

Swiftradish uses the Next.js App Router partitioned into four route groups: (public), (auth), (main), and (legal). On **Desktop Viewports (\ge 1280\text{px})**, the interface renders a three-column layout featuring a fixed left navigation rail (260px width housing Home /home, Explore & Patches /explore, My Co-Plots /plots, Activity /notifications, Bookmarks /bookmarks, Profile /u/[username], Settings /settings, and the primary CTA button + Plant Post / Co-Plot), a fluid center canvas (640px maximum width), and a sticky right contextual panel (340px width displaying upcoming Co-Plots on /home or the RSVP/Contributor roster and Plot Thread on /p/[postId]), with Stage 2 Messages /messages mounting between Co-Plots and Activity when enabled. On **Tablet Viewports (768\text{px}–1279\text{px})**, the layout adapts to a collapsible 72px icon-only left rail paired with the 640px center canvas, while the contextual right panel collapses into a slide-over drawer. On **Mobile and Installed PWA Viewports (< 768\text{px})**, the interface presents a persistent bottom navigation bar (64px height plus env(safe-area-inset-bottom)) exposing five thumb-zone destinations—**Home**, **Explore**, **+ Plant** (full-screen modal composer), **Co-Plots** (upgraded to the **Messages** tab in Stage 2 while Co-Plots sits in the header and profile), and **Profile**—paired with a contextual top app bar housing **Search** (/search) and **Notifications** (/notifications).
Under the **Deep-Link, History, and Auth Boundary Contract**, navigating to /p/[postId], /u/[username], or /patches/[slug] while unauthenticated renders a complete server-side rendered (SSR) read-only view (SCR-02, SCR-03) with dynamic OpenGraph meta tags (og:title, og:description, og:image). Clicking any mutation control (React, Join Plot, Submit Photo, Comment, Follow) opens the Auth Modal or redirects to /login?next=<encoded_internal_path>, where next is strictly validated to start with a single / (and never // or http) to prevent open-redirect vulnerabilities. Closing a modal lightbox (?media=2) invokes router.back() to restore the exact scroll offset of the underlying feed.

4.2 Audit of the 40-Screen Catalog and MVP Screen/State Coverage Matrix

Auditing the prior blueprint's 40 screens (SCR-01 to SCR-40) confirms that the screen IDs are well-structured and should be preserved to avoid numbering drift, but **22 screens form the**

Mandatory MVP Launch Core (Slices 1–6) while **18 screens are explicitly staged into Stage 2 Fast-Follow (Slices 7–9)** so the solo builder can ship a working, end-to-end product incrementally.

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
SCR-01: Public Landing & Manifesto (/)	**MVP Core (Slice 1)**	Understand product thesis; preview live public Co-Plots; sign up.	Primary: Join Swiftradish (/register), Sign In (/login). Secondary: Browse public Patches (/explore).	GET /api/v1/public/showcase (SSR cached 60s). Client state: local interactive Co-Plot preview toggle.	Public (anon). Redirects 307 to /home if valid session cookie exists.	**Load:** Static SSR shell (0 spinners). **Err/Timeout:** Fallback to static curated showcase card.	Landmarks: <header>, <main>, <footer>. Contrast \ge 4.5:1. Event: landing_cta_clicked.	LCP \le 1.2\text{s}; zero fabricated user counters; interactive Co-Plot demo card toggles locally.
SCR-02: Public Post & Co-Plot Reader (/p/[postId])	**MVP Core (Slice 4)**	View shared post/Co-Plot from external link (WhatsApp/X).	Primary: Swipe carousel, read RSVP count & comments. Secondary: Click Join Plot -> triggers Auth prompt.	GET /api/v1/posts/{id} (SSR + OpenGraph metadata). Server-authoritative visibility check.	Public if post.visibility == 'PUBLIC' and author is_private == FALSE; else returns 404.	**Load:** Aspect-ratio-locked Blurhash skeleton. **Empty/404:** Pip 404 card ("Plot not found or private").	Full keyboard arrow navigation on carousel; alt text rendered on all images. Event: public_post_viewed.	Serves valid og:image absolute URL; mutation clicks open SCR-04 preserving ?next=/p/{id}.
SCR-03: Public Patch Showcase (/patches/[slug])	**MVP Core (Slice 5)**	Preview a topic community before registering.	Primary: Read public Patch posts and upcoming	GET /api/v1/patches/{slug}, GET /api/v1/patches/{slug	Public read-only for public posts; hides member	**Load:** Header + 2-column grid skeleton. **Empty:**	<h1> Patch title; semantic feed list (role="fee	Displays accurate member count and rules; blocks

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			Co-Plots. Secondary: Join Patch (triggers Auth).	}/posts (public subset).	roster from unauthenticated scrapers.	"Freshly planted Patch—sign in to add the first post".	d"). Event: public_patch_viewed.	member directory scraping for anon callers.
SCR-04: Sign-Up, 18+ Age Gate & Consent (/register)	**MVP Core (Slice 1)**	Create an 18+ account with explicit DPDP consent notice.	Primary: Submit DOB, email, password, and itemized consent checkbox; or Google OAuth.	Supabase Auth sign-up + POST /api/v1/auth/register. Local state: form validation & DOB check.	Public (anon). Rejects any DOB calculating to < 18 years old.	**Disabled:** Submit disabled until valid fields + consent checked. **Err:** Generic message on existing email; **Age Reject:** Non-retriable 18+ policy screen.	aria-invalid and aria-describedby linked to inline field errors; keyboard DOB inputs. Event: signup_submitted.	Rejects < 18 DOB with HTTP 422; sets HttpOnly; Secure; SameSite=Lax cookies; zero personal data stored on rejected underage attempt.
SCR-05: Login Surface (/login)	**MVP Core (Slice 1)**	Authenticate returning user and restore deep link.	Primary: Sign in via email/password or OAuth. Secondary: Password visibility toggle, /auth/rese	Supabase Auth signInWithPassword + GET /api/v1/users/me.	Public (anon). Rate-limited to 5 failed attempts per 15 min per IP/email.	**Load:** Button spinner, inputs disabled. **Err:** Invalid credentials banner; **429:** Cooldown	Focus moves automatically to error banner (role="alert") on failure. Event: login_suc	Restores internal next route safely; rejects external next=https://evil.com redirects.

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
			t-password.			timer banner.	ceeded.	
SCR-06: Password Recovery & Reset (/auth/reset-password)	**MVP Core (Slice 1)**	Request recovery link and set new password.	Step 1: Submit email. Step 2: Enter new password + confirmation.	Supabase Auth recovery flow + session revocation (POST /api/v1/auth/revoke-all).	Public (anon) for Step 1; single-use recovery token required for Step 2.	**Success:** Identical confirmation message whether email exists or not. **Err:** Expired token with 1-tap Resend.	Live password strength indicator announced via aria-live="polite". Event: password_reset_completed.	Zero email enumeration timing/copy difference; revokes prior refresh tokens on password change.
SCR-07: Email Verification Gate (/auth/verify)	**MVP Core (Slice 1)**	Verify email via 6-digit OTP or magic link.	Primary: Enter/paste 6-digit OTP. Secondary: Resend code (60s cooldown timer).	POST /api/v1/auth/verify-otp (via Supabase Auth OTP verification).	Authenticated session with email_verified == FALSE.	**Load:** Verifying indicator. **Err:** Invalid OTP shake; locks for 15 min after 5 wrong guesses.	6 inputs grouped in <fieldset> with paste auto-fill across boxes. Event: email_verified.	Supports full 6-digit clipboard paste; advances immediately to SCR-08 upon verification.
SCR-08: Onboarding Step 1 — Handle & Identity (/onboarding/profile)	**MVP Core (Slice 1)**	Claim @username, display name, bio, and optional avatar.	Primary: Claim handle and save profile. Secondary: Upload/crop avatar	GET /api/v1/users/check-username?q=... (300ms debounce), PATCH /api/v1/us	Authenticated (onboarding_step == 1).	**Load:** Debounced availability spinner. **Conflict Err:** Suggests 3	Live availability status via aria-live="polite". Event: onboarding_profile_saved.	Enforces ^[a-z0-9_]{3,24}$; avatar uploads via presigned PUT to B2 and strips

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
			image.	ers/me.		available handle variants.		EXIF.
SCR-09: Onboarding Step 2 — Patch Seeding (/onboarding/interests)	**MVP Core (Slice 5)**	Join \ge 3 starter Radish Patches to populate initial feeds.	Primary: Toggle starter Patches and click Enter Swiftradish. Secondary: Skip to blank feed.	GET /api/v1/onboarding/starter-patches, POST /api/v1/onboarding/complete.	Authenticated (onboarding_step == 2).	**Partial:** Counter shows 1 of 3 selected. **Success:** Unlocks "First Sprout" badge and routes to /home.	Patch cards use role="checkbox" and aria-checked, keyboard Space/Enter toggleable.	Joining 3 Patches inserts patch_members rows atomically and populates /home immediately.
SCR-11: Home Feed — "My Patch" Chronological (/home?tab=following)	**MVP Core (Slice 5)**	Catch up on posts and Co-Plots from followed users and joined Patches.	Primary: Scroll feed, react, comment, RSVP to Co-Plots. Secondary: Switch tab to Fresh Harvest.	GET /api/v1/feed/following?cursor=... (TanStack Query infinite keyset pagination).	Authenticated (onboarding_completed == TRUE).	**Load:** 3 structural card skeletons. **Empty:** Honest onboarding card linking to /explore. **Offline:** Cached feed banner + Retry.	Feed cards navigable via J/K shortcuts or Tab; images carry explicit width/height + alt.	Strictly ordered by id DESC (UUIDv7 chronological); renders "All caught up" divider at last_seen_post_id.
SCR-12: Home Feed — "Fresh Harvest"	**MVP Core (Slice 5)**	Discover relevant public posts and Co-Plots	Primary: Browse discovery feed; click provenan	GET /api/v1/feed/discover?cursor=..., POST	Authenticated.	**Empty:** Falls back to all-time featured public	Provenance popover accessible via	Every card displays an honest provenan

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
Discovery (/home?tab=discover)		with transparent ranking.	ce pill (Why this post?). Secondary: Mute topic or Mute author.	/api/v1/safety/mutes.		Patches if filters are too narrow. **Err:** Inline retry card.	keyboard button (aria-expanded). Event: feed_provenance_inspected.	ce reason badge and 1-tap topic/author mute action.
SCR-13: Post & Co-Plot Detail View (/p/[postId])	**MVP Core (Slice 4)**	Inspect full carousel, RSVP to Co-Plot, submit shared slide, and join discussion.	Primary: Going/Maybe RSVP toggle, + Add Photo to Plot, post comment/reply. Secondary: Bookmark, Share, Report.	GET /api/v1/posts/{id}, GET /api/v1/posts/{id}/comments, POST /api/v1/plots/{id}/rsvp, WebSocket room subscription for live Plot thread.	Authenticated (respects PUBLIC, FOLLOWERS, MUTUALS, PATCH_ONLY and block lists).	**Optimistic:** Transition from feed cache while comments load. **Partial Err:** Comment tree error shows inline retry without hiding post.	Deep-links to #comment-{id} with focus management; RSVP buttons use aria-pressed.	Enforces 3-level max comment depth (0..2); RSVP updates roster and unlocks Plot discussion immediately.
SCR-14: Media Lightbox & Carousel Inspector (/p/[postId]?media=[index])	**MVP Core (Slice 3)**	View full-resolution 2048w media, contributor credit, and full alt-text.	Primary: Next/Prev slide, zoom/pan, toggle full alt-text overlay, close modal.	Hydrated from parent post.media_assets array.	Matches parent post visibility.	**Load:** Blurhash canvas cross-fades to full_2048w.webp. **Err:** Broken image fallback	Strict focus trap (Radix Dialog); Escape closes and restores trigger focus; Left/Right	Zero background page scroll bleed while open; displays @contributor credit badge on

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
						with Reload image.	arrow keys switch slides.	contributed Co-Plot slides.
SCR-15: Unified Post & Co-Plot Composer (/compose or Modal)	**MVP Core (Slice 3 & 4)**	Publish rich Markdown-lite text, 1–10 image carousels, and interactive Co-Plots.	Primary: Attach up to 10 images, toggle Make Co-Plot (set title, date/time, location, allow shared uploads), select visibility, click Plant Post.	POST /api/v1/media/upload-intents, Presigned PUT to B2, POST /api/v1/media/{id}/complete, POST /api/v1/posts, IndexedDB local draft.	Authenticated active user.	**8 Deterministic States:** EMPTY -> VALIDATING -> INTENT_AUTHORIZED -> UPLOADING -> VERIFYING -> UPLOAD_FAILED (per-file Retry/Remove) -> MEDIA_READY -> POST_COMMITTED.	Progress rings expose role="progressbar" + aria-valuenow; per-file cancel/retry keyboard reachable. Event: post_published.	Plant Post remains disabled until all attached tiles reach READY; resets <input type="file"> and revokes blob URLs on commit.
SCR-16: Multi-Image Crop, Reorder & Alt-Text Drawer (Sub-view of	**MVP Core (Slice 3)**	Crop aspect ratios (1:1, 4:5, 16:9), reorder carousel slides,	Primary: Apply crop, move slide Left/Right (or drag), enter	Client-side HTML5 Canvas / WebWorker (strips EXIF client-side before	Authenticated.	**Processing:** WebWorker crop spinner (< 200\text{ms}). **Cancel:**	Keyboard-accessible Move Left / Move Right buttons alongside	Client strips EXIF GPS tags prior to upload; server re-verifies and strips

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
SCR-15)		and enter alt-text.	alt-text (up to 1,000 chars), Save Changes.	presigned PUT).		Restores original uncropped selection.	drag-and-drop so non-pointer users can reorder slides.	metadata via pyvips.
SCR-17: Co-Plot Contribution Approval Queue (/p/[postId]/manage)	**MVP Core (Slice 4)**	Allow Co-Plot host to review and approve/decline participant photo submissions.	Primary: Approve to Carousel or Decline per pending slide. Secondary: Toggle auto-approve for mutuals.	GET /api/v1/plots/{id}/submissions, POST /api/v1/plots/{id}/submissions/{subId}/decide.	Strictly restricted to post.author_id (returns 404/403 to all others).	**Empty:** Pip relaxing ("No pending slide submissions"). **Destructive:** Decline prompts confirmation or 5s undo toast.	Action buttons clearly labeled (Approve photo from @user); status changes announced to screen readers.	Approving a slide atomically appends it to parent post carousel (up to 25 total collaborative slides) and notifies contributor.
SCR-19 & SCR-20: Explore Directory & Unified Search (/explore, /search)	**MVP Core (Slice 5)**	Discover featured Patches, weekend Co-Plots, and search @users, #tags, Patches, and posts.	Primary: Type query in command bar (Cmd/Ctrl+K), switch result tabs (All, People, Co-Plots & Posts,	GET /api/v1/explore/hub, GET /api/v1/search?q=...&type=...&cursor=.. . (250ms debounce + AbortController).	Authenticated (filters out blocked/muted accounts and non-public posts).	**Load:** Shimmer bar + grid skeletons. **Empty:** Spelling suggestions via pg_trgm similarity. **Err:** Retry button.	Combobox pattern (role="combobox", aria-activedescendant) for typeahead dropdown. Event: search_executed.	Aborts stale in-flight requests on rapid typing; executes weighted tsvector + pg_trgm query in \le 25\text{m

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
			Patches), join Patches.					s} p95.
SCR-21: Radish Patch Community Space (/patches/[slug])	**MVP Core (Slice 5)**	Participate in a topic community's curated feed and active Co-Plots.	Primary: Join Patch, + Plant in Patch, switch between Curated Feed and Patch Co-Plots tabs.	GET /api/v1/patches/{slug}, POST /api/v1/patches/{slug}/join, GET /api/v1/patches/{slug}/posts.	Public Patches readable by all; posting requires active row in patch_members.	**Optimistic:** Join button updates immediately. **Empty:** Prompt to plant the first post in the Patch.	Tab list uses role="tablist" and role="tab" with arrow-key switching. Event: patch_joined.	Joining a Patch immediately includes its posts in the user's My Patch home feed.
SCR-23 & SCR-24: User Profile & Profile Editor (/u/[username], /settings/profile)	**MVP Core (Slice 2)**	Express identity, view user's Posts and Co-Plots, follow/unfollow, and edit profile/theme.	Primary: Follow / Request Follow / Mutual button, switch Posts and Co-Plots tabs, edit bio/avatar/theme swatch.	GET /api/v1/users/{username}/profile, GET /api/v1/users/{username}/posts, PATCH /api/v1/users/me/profile.	Enforces is_private and user_blocks server-side (blocked callers receive minimal shell or 404).	**Private State:** "This profile is private—send a follow request to collaborate." **Success:** Live preview and toast on profile save.	Theme swatches expose explicit text labels (Radish Crimson, Sprout Lime, etc.) and guarantee \ge 4.5:1 contrast.	Private accounts hide posts/followers from non-followers at the SQL query layer, not merely in CSS.
SCR-25: Relationships Manager (/u/[usern	**MVP Core (Slice 2)**	Manage Followers, Following, Mutuals, and	Primary: Approve or Decline follow requests;	GET /api/v1/users/{id}/relations?tab=...,	Requests tab strictly restricted to profile owner;	**Destructive:** Unfollowing a private	Accessible list items with distinct button	Approving a reciprocal follow triggers

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
ame]/relations)		incoming Follow Requests.	Unfollow or Remove Follower.	POST /api/v1/relations/requests/{id}/decide.	other tabs respect is_private.	account prompts confirmation dialog warning that re-following requires approval.	labels (Approve @handle, Decline @handle).	database is_mutual = TRUE on both directional edges atomically.
SCR-31: Activity Center — Notifications Hub (/notifications)	**MVP Core (Slice 6)**	Review deduplicated updates on Co-Plots, RSVPs, slide submissions, follows, and replies.	Primary: Filter by All, Co-Plots & RSVPs, Mentions & Replies; click row to deep-link; Mark all read.	GET /api/v1/notifications?filter=..., POST /api/v1/notifications/mark-read.	Authenticated (recipient only).	**Empty:** Pip sleeping ("All quiet in the patch"). **Unread:** Crimson dot indicator (#FF4D5E) + bold actor summary.	Unread state conveyed via text (Unread:) in addition to crimson dot. Event: notification_opened.	High-frequency reactions roll up via dedupe_key (@alice and 4 others reacted); clicking deep-links to exact context.
SCR-33 & SCR-34: Settings — Account, Security, Privacy & Blocks (/settings/account, /settings/privacy)	**MVP Core (Slice 6)**	Manage sessions, toggle private account, manage blocked/muted users, export or delete account.	Primary: Revoke Session, toggle Private Account, Unblock/Unmute, Export My Data, Delete Account (SCR-37	GET/DELETE /api/v1/auth/sessions, PATCH /api/v1/users/me/privacy, GET/DELETE /api/v1/safety/blocks, POST	Authenticated; account deletion requires step-up credential verification and typing DELETE.	**Confirm Dialog:** Switching Private -> Public warns that pending follow requests auto-approve; deletion explains	All toggle switches use role="switch" with aria-checked; destructive dialogs focus Cancel button by default.	Revoking a session writes its JWT jti to Redis denylist; account deletion hides profile immediately and schedules

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
			modal).	/api/v1/account/schedule-deletion.		14-day grace window.		B2 media purge.
SCR-36, SCR-38 & SCR-39: Statutory Report Modal, User Ticket Tracker & Admin Triage Console (/settings/reports, /internal/moderation)	**MVP Core (Slice 6)**	File abuse/statutory reports, track SLA tickets, appeal takedowns, and resolve moderation queue.	Primary: Submit report (SCR-36) with optional 1-tap Block; view ticket SLA status & file appeal (SCR-38); Admin resolve (Remove, Suspend, Dismiss) (SCR-39).	POST /api/v1/safety/reports, GET /api/v1/safety/my-reports, POST /api/v1/safety/appeals, GET/POST /api/v1/admin/moderation/....	SCR-36/38 authenticated (plus /legal/grievance web form for non-users); SCR-39 restricted to role IN ('admin', 'grievance_officer').	**Immediate Shield:** Reported item immediately collapses into muted shield in reporter's viewport. **SLA Alert:** CRITICAL_24H items highlighted in crimson at top of SCR-39.	Wizard steps announce progress (Step 1 of 3); ticket ID copy button announces Copied to clipboard.	Generates ticket ID and acknowledgment in < 60\text{s}; enforces 24h (Rule 3(2)(b)), 72h (Rule 3(1)(b)), and 15d SLAs; logs immutable audit_events row.
SCR-40: Global Error, Offline, 404 & 429 Surfaces (not-found.t[span_108](start_span)[spa	**MVP Core (Slice 1)**	Provide honest recovery paths during network loss, missing routes, or rate	Primary: Retry Request, Return Home, or read cached offline content.	Service Worker cache for /offline; Retry-After header parsing for HTTP 429.	Public and Authenticated.	**429 State:** Live countdown timer derived from Retry-After header before	Error heading receives focus on mount; countdown updates throttled for screen readers.	Never exposes raw Python stack traces or SQL errors in production.

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
n_108](end_span)[span_112](start_span)[span_112](end_span)sx, error.tsx, /offline)		limiting.				enabling Retry button.		
Stage 2 Screens (SCR-10, 18, 22, 26–30, 32, 35)	**Stage 2 Fast-Follow (Slices 7–9)**	Expand into standalone 1:1/group DM Inbox (SCR-27–30), Co-Plot Calendar & .ics export (SCR-18), Patch Creation & Steward Console (SCR-22), Bookmark Folders (SCR-26), PWA Push Primer & Quiet Hours (SCR-10,	Full specifications preserved from Blueprint Document C (SCR-10 through SCR-35).	Uses same normalized tables (conversations, messages, bookmarks, patches, push_subscriptions).	Enforces two-plane RBAC + Object Policy across all Stage 2 routes.	Full loading, empty, optimistic, and reconnect states as specified in Section I.	Full WCAG 2.2 AA compliance required at each Stage 2 slice gate.	Stage 2 slices unlock only after MVP Core Slices 1–6 pass end-to-end integration and browser verification.

Screen ID & Stable Route	Stage / Slice	User Goal & Entry Points	Primary & Secondary Actions	Required API & State Dependencies	Permissions & Visibility Rules	Comprehensive UI States (Load, Empty, Partial, Err, Offline, Retry, Destructive)	Accessibility (WCAG 2.2 AA) & Analytics Events	Measurable Acceptance Criteria
		SCR-35), and Badge Showcase (SCR-32).						

E. Canonical User Journeys and State Machines

Rather than relying on informal descriptions, all critical user flows are governed by deterministic client and server state transitions that explicitly separate temporary client state (Zustand / IndexedDB / React state) from durable server-authoritative state (PostgreSQL 16 + Backblaze B2 + Redis 7).

5.1 Journey 1: Registration, 18+ Age Gate, Consent, and Cold-Start Onboarding

When an unauthenticated user navigates to /register (SCR-04) or clicks Join Plot on public SCR-02 (which sets ?next=/p/{postId}), client state initializes at AUTH_IDLE. As the user enters their Date of Birth, email, and password and checks the standalone DPDP consent notice, the flow enters AUTH_VALIDATING. If the calculated age is under 18, the client transitions immediately to AGE_REJECTED—and the server rejects any bypassed POST /api/v1/auth/register call with HTTP 422 ERR_AGE_RESTRICTED—ensuring zero personal data is written to PostgreSQL or Supabase Auth. If the submitted email already exists, the server returns a generic verification prompt (AWAITING_OTP) with an identical response timing floor (250\text{ms} minimum) so attackers cannot enumerate registered email addresses, while exceeding 5 attempts in 15 minutes returns HTTP 429 with a Retry-After header.
During OTP verification (SCR-07, transitioning from AWAITING_OTP to EMAIL_VERIFIED), the user submits the 6-digit OTP sent by Supabase Auth (POST /api/v1/auth/verify-otp). On success, Supabase sets HttpOnly; Secure; SameSite=Lax session cookies, FastAPI verifies the ES256 JWT via cached JWKS, and the backend inserts the durable users (onboarding_step = 1) and user_profiles rows inside a single PostgreSQL transaction. In the subsequent handle claim step (SCR-08, PROFILE_SETUP), the client debounces GET /api/v1/users/check-username?q={handle} by 300ms; if the handle is taken, the server returns three available suffixed or prefixed suggestions, and submitting PATCH /api/v1/users/me commits the unique username (UNIQUE LOWER(username)) and advances onboarding_step = 2. Finally, during cold-start Patch seeding (SCR-09, INTEREST_SEEDING to ONBOARDING_COMPLETE), the user selects at least three starter Radish Patches (POST

/api/v1/onboarding/complete), which atomically inserts patch_members rows, sets onboarding_step = 3 (completed), and redirects either to the validated ?next=/p/{postId} deep link or to /home?tab=following (SCR-11) with a populated feed.

5.2 Journey 2: Co-Plot Creation, RSVP, Collaborative Media Contribution, and Approval

To publish a Co-Plot (SCR-15 to SCR-13), a host attaches 1 to 10 images (verified via the 8-state upload lifecycle detailed in Section I), toggles Make Co-Plot, enters title, starts_at (optional UTC timestamp), location_label, capacity (optional), and contribution_policy (MUTUALS_AUTO or HOST_APPROVAL), and submits POST /api/v1/posts with an Idempotency-Key: <uuidv4> header. FastAPI then executes an atomic transaction creating rows in posts (post_type = 'CO_PLOT'), co_plots, conversations (type = 'CO_PLOT_THREAD'), and conversation_participants (adding the host as HOST), linking the uploaded media_assets rows (post_id = new_post.id), and returning HTTP 201.
When a participant clicks Going (POST /api/v1/plots/{postId}/rsvp with {"status": "GOING"}) on SCR-13, the server locks the co_plots row (SELECT ... FOR UPDATE), checks going_count < capacity, upserts co_plot_participants, increments going_count, and inserts the user into conversation_participants for the Plot Thread. If capacity was reached concurrently, the server returns HTTP 409 ERR_PLOT_FULL and the client reverts the optimistic button state with an explanatory toast. Changing an RSVP from GOING to MAYBE or DECLINED decrements going_count atomically, and selecting DECLINED revokes active write access to the Plot Thread while preserving prior messages.
For collaborative carousel contributions (SCR-13 to SCR-17), an authenticated participant taps + Add Photo to Plot on SCR-13, uploads 1 to 3 photos via presigned PUT to Backblaze B2 (MEDIA_READY), and calls POST /api/v1/plots/{postId}/submissions. Under Branch A (Mutual with MUTUALS_AUTO policy), if EXISTS (SELECT 1 FROM user_relationships WHERE actor_id = host AND target_id = contributor AND is_mutual = TRUE), the server immediately sets submission.status = 'APPROVED', assigns the next position on media_assets (post_id = postId, contributor_id = contributor), and emits a notification to the host. Under Branch B (Non-Mutual or HOST_APPROVAL policy), the server sets submission.status = 'PENDING_APPROVAL' and notifies the host, who opens SCR-17 (/p/{postId}/manage) and clicks Approve (POST /api/v1/plots/{postId}/submissions/{subId}/decide) to atomically attach the slide to the carousel and notify the contributor, or Decline to mark the submission REJECTED and schedule the unattached B2 object for cleanup.

5.3 Journey 3: Privacy Changes, Blocking, Reporting, Moderation, and Account Deletion

When User A blocks User B (POST /api/v1/safety/blocks, triggered from the SCR-23 profile menu or inside the SCR-36 report flow), PostgreSQL executes a single atomic transaction that inserts (user_a, user_b) into user_blocks, deletes any user_relationships rows in both directions (A->B and B->A) while decrementing follower and following counters, removes User B from any upcoming Co-Plots hosted by User A, invalidates the Redis block-list caches blocks:{user_a} and blocked_by:{user_b}, and immediately drops any active WebSocket subscription where User B is viewing User A's resources.
In the statutory reporting and takedown SLA workflow (SCR-36 to SCR-39 to SCR-38), a

reporter submits POST /api/v1/safety/reports with category and target_type/target_id, immediately transitioning the item to REPORTED_HIDDEN in local client cache. The backend classifies the statutory SLA tier into CRITICAL_24H (Rule 3(2)(b): bodily privacy violations, non-consensual imagery, or identity morphing/impersonation), STANDARD_72H (Rule 3(1)(b): hate speech, child harm, gender harassment, misinformation, or sovereignty threats), or GENERAL_15D (intellectual property infringement or general grievances). For CRITICAL_24H reports, if an object receives two or more distinct trusted user reports or matches severe safety heuristics, the server places the post in PENDING_MOD_SHIELD (temporarily hiding public rendering while preserving evidence) and pages the Grievance Officer via high-priority alert. Within 60 seconds, the system dispatches an automated in-app and email acknowledgment containing the unique ticket_number (SR-2026-XXXXXX). When the Grievance Officer resolves the case in SCR-39 (Remove Content), the server soft-deletes the post (deleted_at = NOW(), moderation_hold = TRUE), retains an encrypted evidence snapshot and B2 object lock/quarantine record for **180 days** per Rule 3(1)(g), notifies both the reporter and author in SCR-38, and enables the author's in-app dispute form and 30-day GAC appeal notice (gac.gov.in).

For account export and deletion (SCR-33 and SCR-37), requesting an export (POST /api/v1/account/export) enqueues an arq job (rate-limited to 1 request per 24 hours) that compiles the user's profile, posts, comments, messages, and media links into a ZIP archive in the private B2 bucket and emails a 24-hour signed download URL. Requesting account deletion (POST /api/v1/account/schedule-deletion) requires step-up password or OTP verification, immediately sets users.status = 'PENDING_DELETION' and deleted_at = NOW(), hides all profile and post records from public and authenticated queries, revokes all Supabase refresh tokens, adds active JWT jti claims to the Redis denylist, and schedules account.hard_purge for NOW() + 14 days. If the user does not log back in to cancel within 14 days, the arq worker permanently deletes all user B2 media objects (except items held under an active 180-day Rule 3(1)(g) legal investigation hold), anonymizes historical group-chat sender references (sender_id = NULL), and cascades deletion of the users row.

F. Design System and Stitch Execution Pack

6.1 Visual Direction Evaluation and Selected Recommendation

Three distinct aesthetic directions were evaluated against the product thesis ([RECOMMENDATION], High Confidence). **Direction 1 ("Nocturnal Botanical Studio" — Warm Carbon-Slate Dark-First + Daylight Parchment Light Companion)** is selected as the canonical visual identity. It pairs warm obsidian-soil surfaces (#0B0D11 canvas, #12151C primary card surface, #1A1E28 elevated surface) with crisp botanical crimson (#FF4D5E) for primary actions and fresh sprout lime (#2EE59D) exclusively for co-active states (Co-Plot indicator bars, Going RSVPs, and Mutual badges). This palette avoids pure #000000 OLED black (which induces astigmatism halation) and avoids GPU-heavy glassmorphism, relying instead on crisp 1px hairline borders (rgba(255,255,255,0.08)) so multi-image photography and carousels command visual focus. Conversely, **Direction 2 ("Sunlit Farmers' Market" — Warm Editorial Serif Light-Only)** was rejected as the primary theme because parchment backgrounds (#FAF7F2) with Fraunces serif headings feel like a static recipe newsletter rather than a live evening coordination platform, though its high-contrast light surface rules inform the companion **Daylight Parchment** light theme (#F6F7F9 canvas, #FFFFFF cards) for outdoor

glare readability. **Direction 3 ("Neon Hyper-Glass" — Pure Black + Translucent Blur Gradients)** was rejected because heavy backdrop-filter: blur(24px) cards degrade mobile scrolling frame rates on mid-tier Android devices, fail WCAG contrast over user photos, and resemble generic AI SaaS templates.

6.2 Canonical Design Tokens (DESIGN.md & globals.css)

Typography across Swiftradish is governed by three semantic font stacks. Display and heading elements (--font-display) use Plus Jakarta Sans (weights 600 and 700, tracking -0.02em), scaling from H1 (clamp(1.75rem, 2.5vw, 2.25rem) at 1.2 line-height) to H2 (clamp(1.35rem, 2vw, 1.65rem) at 1.25) and H3 (1.125rem / 18px at 1.35). UI controls and body copy (--font-sans) use Inter (weights 400, 500, 600), setting primary body text at 1rem (16px) with a 1.55 line-height (max-width: 65ch), compact comment text at 0.9375rem (15px, 1.5 line-height), and secondary captions at 0.8125rem (13px, 1.4 line-height). Numeric counters, SLA timers, and timestamps (--font-mono) use JetBrains Mono (0.75rem–0.8125rem, font-variant-numeric: tabular-nums) so live RSVP updates (12/15 Going) never cause horizontal layout jitter.
Spatial layout follows a base 4px (0.25rem) grid (4, 8, 12, 16, 24, 32, 48, 64px), with feed cards using 16px internal padding on mobile, 20px on desktop, and a 16px vertical gap between cards. Border radii are standardized at 14px for card containers, 10px for internal media stages, buttons, and inputs, and 9999px for status and topic pills. Elevation and z-index layering follow a strict six-tier hierarchy: base canvas at 0, sticky feed header and bottom navigation at 20, dropdowns and popovers at 30, slide-over drawers (SCR-17, SCR-30) at 40, modal composer and lightbox surfaces (SCR-14, SCR-15, SCR-36) at 50, and toast notifications at 60.
In Dark Mode ([data-theme="dark"], default), semantic color tokens map to --bg-canvas: #0B0D11, --bg-surface-1: #12151C, --bg-surface-2: #1A1E28, --bg-surface-3: #232936, --border-subtle: rgba(255,255,255,0.08), --text-primary: #F3F5F8 (16.4:1 contrast on Surface 1), --text-secondary: #9BA3B4 (6.8:1 contrast on Surface 1), --accent-primary: #FF4D5E (paired with #0B0D11 dark ink text on solid primary buttons for 7.9:1 contrast), --accent-secondary: #2EE59D (paired with #0B0D11 text on solid badges), --status-success: #10B981, --status-warn: #F59E0B, and --status-error: #EF4444. In Light Mode ([data-theme="light"]), tokens map to --bg-canvas: #F6F7F9, --bg-surface-1: #FFFFFF, --bg-surface-2: #EEF0F4, --bg-surface-3: #E2E6ED, --border-subtle: rgba(15,23,42,0.09), --text-primary: #0F141C (17.1:1), --text-secondary: #4B5565 (6.4:1), --accent-primary: #E02E40 (paired with #FFFFFF text for 4.6:1 contrast), and --accent-secondary: #0D9F6E. Every interactive primitive enforces a :focus-visible ring of 2px solid var(--accent-secondary) with a 2px offset.

6.3 Core Component Contracts and Anti-Bloat Rules

Standard post cards render the author header (avatar, display name, @handle, Patch pill, relative timestamp), Markdown-lite body copy (collapsed after 6 lines with Read more), a 1-to-10-slide aspect-ratio-locked media carousel (4:5 portrait, 1:1 square, or 16:9 landscape—preventing layout shift during load via server-stored dimensions and a Blurhash canvas), and the action bar (React, Comment count, Bookmark, Share). **Co-Plot cards** add a 3px solid --accent-secondary (#2EE59D) left accent border and an embedded **Co-Plot Coordination Tray** beneath the media stage displaying the activity title, JetBrains Mono date/time and location pill, an avatar stack of Going participants (8/12 Going), a 1-tap Join Plot button, and a + Add Slide pill when collaborative carousel uploads are enabled. To prevent illustration production from blocking core engineering, **Pip the Radish** is restricted in the MVP

Core to **8 clean layered SVG files** (< 8KB gzipped each, stored in frontend/public/pip/core/) covering Onboarding (pip_sprout_joyful), Empty Feed/Search (pip_sprout_curious), Upload Inspecting (pip_creative_active), Upload/Form Error (pip_system_oops), Offline/404 (pip_system_chill), Safety Shield (pip_safety_focused), and First Co-Plot Celebration (pip_coplot_hype), deferring the 30-pose and 216-pose vector rig expansions until post-launch.

6.4 Google Stitch Execution Pack (Prompts, Batches, and Rejection Gates)

In accordance with project constraints, Google Stitch (https://stitch.googleapis.com/mcp, 350 free generations/month) is used **only as a visual exploration and layout reference tool**, never as a source of application logic or unreviewed code ([USER DECISION]).

Canonical Design-System Context Prompt (Committed to DESIGN.md)

"Design a high-craft, dark-first co-active social web application named Swiftradish using the 'Nocturnal Botanical Studio' system. Viewport backgrounds use warm obsidian #0B0D11, primary cards use #12151C, elevated drawers/modals use #1A1E28, and structural dividers use 1px solid rgba(255,255,255,0.08) with zero neon glow gradients and zero heavy drop shadows. Primary text is #F3F5F8 (Plus Jakarta Sans 600/700 for headings, Inter 400/500 for body), secondary metadata is #9BA3B4, and numeric counters/timestamps use tabular JetBrains Mono. Primary CTA buttons use botanical crimson #FF4D5E with dark #0B0D11 label text; co-active Co-Plot status bars, mutual badges, and 'Going' RSVP pills use sprout lime #2EE59D. Desktop uses a 3-column layout (260px left nav rail, 640px max center feed canvas, 340px right contextual panel); mobile (390x844) uses a 64px 5-tab bottom navigation bar and top contextual header. Maintain realistic content density: max 4 data points per card header, generous 16px–20px padding, and minimum 44x44px touch targets."

Anchor Screen Prompts and Batch Execution Protocol

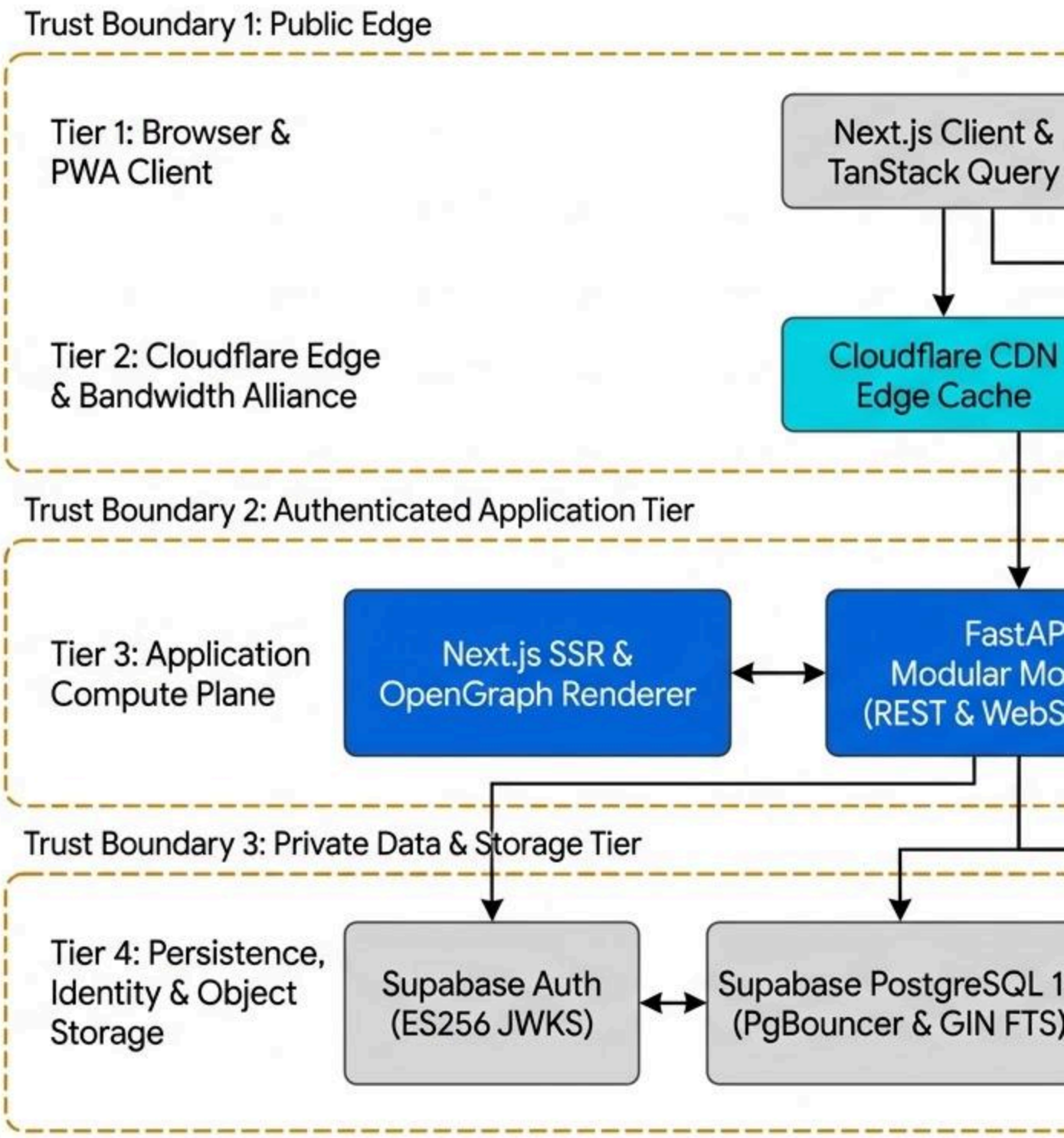
Before generating secondary views, the solo builder executes **Batch 0 (Anchor Screens)** in Stitch and locks the extracted design tokens. For **Anchor 1 (SCR-11 Home Feed /home)**, the prompt specifies Desktop (1440x900) and Mobile (390x844) views where the left rail displays the Swiftradish header, Home (active), Explore & Patches, Co-Plots, Activity (with badge 3), Bookmarks, Profile, and a crimson + Plant Post / Plot CTA; the center canvas renders the sticky tab switcher (My Patch — Following active, Fresh Harvest — Discover inactive), an All Caught Up divider, one Co-Plot card (with #2EE59D left accent bar, 3-photo carousel from an urban sketching walk, Co-Plot tray showing Saturday 4:00 PM • Cubbon Park • 9/12 Going, Going active pill, and + Add Photo button), and one standard post card; and the right panel displays Upcoming Co-Plots This Weekend and Your Radish Patches. For **Anchor 2 (SCR-13 Post & Co-Plot Detail /p/[postId])**, the prompt specifies a center stage with a 5-slide collaborative photo carousel showing a contributor credit pill (Slide 3 by @arjun_k) and alt-text badge, followed by the Co-Plot Coordination Block (Going: 9, Maybe: 3, + Submit Photo to Carousel button) and a 3-level nested comment and coordination thread with a sticky reply composer, paired with a right panel showing the participant RSVP roster and a host-only banner (2 pending photo submissions — Review). For **Anchor 3 (SCR-15 Unified Post & Co-Plot Composer /compose)**, the prompt specifies a modal composer on #1A1E28 surface (600px width) with a

top segmented toggle (Standard Post vs Co-Plot Activity active), a Markdown-lite text area, a 4-item horizontal media upload tray showing 2 tiles with green #10B981 Ready checkmarks, 1 tile at 68% circular progress ring, and 1 failed tile with #EF4444 border and inline Retry / Remove buttons, followed by Co-Plot fields (Activity Title, Date/Time picker, Location label, and toggle Allow mutuals to add carousel photos) and a footer visibility selector (Public, Patch Only, Mutuals) with a disabled Uploading media... submit button.
Once Batch 0 passes visual inspection and its design context is locked via extract_design_context, remaining screens are generated only as needed across **Batch 1 (Slice 1 & 2):** Auth, Onboarding & Profiles (SCR-01, SCR-04, SCR-05, SCR-08, SCR-23, SCR-25); **Batch 2 (Slice 3 & 4):** Lightbox, Crop/Alt-Text & Co-Plot Approval (SCR-02, SCR-14, SCR-16, SCR-17); **Batch 3 (Slice 5):** Discovery, Search & Patches (SCR-03, SCR-09, SCR-12, SCR-19, SCR-20, SCR-21); **Batch 4 (Slice 6):** Notifications, Settings, Safety & Triage (SCR-31, SCR-33, SCR-34, SCR-36, SCR-38, SCR-39, SCR-40); and **Batch 5 (Stage 2 Deferred):** SCR-10, SCR-18, SCR-22, SCR-26–30, SCR-32, SCR-35. Any generated Stitch screen must be rejected or stripped during React refactoring if it invents unbacked features (such as voice call buttons, stories rings, algorithmic view counts, or crypto wallets), violates WCAG 2.2 AA contrast (< 4.5:1), omits error/empty/loading variants, or embeds hardcoded static DOM trees without Radix UI accessibility primitives (Dialog, DropdownMenu, Tabs, Switch).

G. Technical Architecture and Service Decisions

7.1 Architectural Topology and Reconciled Stack

Swiftradish employs a **Modular Monolith Backend (Python 3.12+ / FastAPI 0.137+ / Starlette 1.0+)** paired with a **Next.js App Router (TypeScript Strict) Frontend**, backed by the user's existing **Supabase Project (PostgreSQL 16 + PgBouncer + Supabase Auth)**, the user's existing **Backblaze B2 Bucket fronted by Cloudflare CDN/Worker (Bandwidth Alliance)**, and **Redis 7 (ARQ Async Workers + WebSocket Pub/Sub + Rate Limiting)** ([RECOMMENDATION], High Confidence).



7.2 Service-by-Service Evaluation, Ownership, and Failure/Exit Contracts

Infrastructure Component	Selected Service & Reconciled Role	Exact Data Owned vs. Not Owned	Plan / Limit / Pricing Assumptions & Primary Source	Security Boundary & Failure / Fallback Behavior	Local Dev Strategy & Exit / Migration Path
1. Frontend & SSR Shell	**Next.js App Router (TypeScript Strict, Tailwind CSS, Radix UI, TanStack Query, Zustand)**. *Alternative evaluated:* Vite SPA (rejected due to lack of SSR OpenGraph unfurling on /p/[id]); FastAPI + HTMX (rejected due to complex multi-image crop/upload and offline IndexedDB state).	**Owns:** SSR HTML rendering, dynamic OpenGraph tags, client routing, Zustand composer state, IndexedDB draft/outbox cache. **Does NOT Own:** Database connections, business rules, or WebSocket termination.	Vercel Hobby/Pro (0–20/mo) or single Docker container behind Cloudflare (5–10/mo).	Never holds database service-role keys. If FastAPI times out (>3s) during SSR of /, falls back to static cached manifesto shell (SCR-01).	Local pnpm dev. Standard React components allow migrating from Vercel to any Node/Docker host in < 1\text{ hour}.
2. Backend API & Realtime Server	**Python 3.12+ / FastAPI 0.137+ (Starlette 1.0+ lifespan) Modular Monolith**. *Alternative evaluated:* Next.js Route Handlers (rejected: serverless functions cannot hold stateful WebSockets);	**Owns:** All REST endpoints (/api/v1/*), Two-Plane authorization enforcement, stateful WebSocket gateway (/ws), and domain service orchestration. **Does NOT Own:** Direct file storage or raw password	Railway / Fly.io / Render container: 1–2x 1GB–2GB RAM instances (7–14/mo at seed launch).	Validates ES256 JWTs locally in < 0.5\text{ms}. Enforces SQL timeouts (5000ms). If WebSockets drop during deploy, clients exponential-backoff and sync via REST/cursor.	Local uvicorn src.main:app --reload. Standard OCI Dockerfile runs on any cloud or VPS without vendor lock-in.

Infrastructure Component	Selected Service & Reconciled Role	Exact Data Owned vs. Not Owned	Plan / Limit / Pricing Assumptions & Primary Source	Security Boundary & Failure / Fallback Behavior	Local Dev Strategy & Exit / Migration Path
	Go/NestJS (rejected: user prefers Python/FastAPI).	hashes.			
3. Primary Database & Search	**User's Existing Supabase PostgreSQL 16 Project (SQLAlchemy 2.0 Async + asyncpg + Alembic)**. *Alternative evaluated:* Separate Neon/RDS instance + Meilisearch (rejected: adds unnecessary vendor sprawl when user already has Supabase Postgres).	**Owns:** All 14 normalized relational tables, UUIDv7 keys, transactional counters, and tsvector + pg_trgm GIN search indexes. **Does NOT Own:** Binary media blobs or ephemeral typing pulses.	Supabase Free Tier ($0/mo, 500MB DB, pauses after 7d inactivity) for initial dev -> **Supabase Pro ($25/mo, 8GB DB included, $0.125/GB overage)** required before live user launch ([VERIFIED]).	Accessed from FastAPI via PgBouncer Transaction Pooler (port 6543, prepared_statement_cache_size=0 in asyncpg) using a dedicated least-privilege application DB role, never exposing PostgREST publicly.	Local postgres:16-alpine in Docker Compose. Pure SQL + Alembic migrations allow pg_dump / replication to AWS RDS, Neon, or self-hosted Postgres at any time.
4. Identity & Authentication	**User's Existing Supabase Auth + FastAPI Local PyJWKClient Verification**. *Alternative evaluated:* Clerk/Auth0 (expensive above 10k MAU); Roll-your-own auth (security	**Owns:** auth.users credentials, OAuth handshakes, OTP generation, refresh token rotation, and asymmetric JWT signing (ES256). **Does NOT Own:** Application RBAC or profile	Included in Supabase Free (50,000 MAU) and Supabase Pro (100,000 MAU included, then $0.00325/MAU) ([VERIFIED]).	FastAPI fetches JWKS from hardcoded https://<project>.supabase.co/auth/v1/.well-known/jwks.json (cached 1h), pins algorithms=["ES256"], verifies aud="authenticated", and checks Redis revoked_jti:{jti}.	Local Supabase Auth container or mock ES256 keypair in pytest. Since users.id matches auth.users.id (UUID), migrating to Ory/Keycloak or native FastAPI OAuth preserves all

Infrastructure Component	Selected Service & Reconciled Role	Exact Data Owned vs. Not Owned	Plan / Limit / Pricing Assumptions & Primary Source	Security Boundary & Failure / Fallback Behavior	Local Dev Strategy & Exit / Migration Path
	risk).	relationships.			foreign keys.
5. Media Object Storage & CDN Delivery	**User's Existing Backblaze B2 Bucket + Cloudflare CDN & Signing Worker (Bandwidth Alliance)**. *Alternative evaluated:* Cloudflare R2 ($0.015/GB/mo) — kept as secondary S3 fallback.	**Owns:** Staging uploads (staging/), processed public WebP variants (media/), private DM/export files (private/), and encrypted DB backups. **Does NOT Own:** Media authorization metadata (owned by Postgres media_assets).	**Backblaze B2:** 10GB free, then **$0.006/GB/mo** ($6/TB); Class A writes free; Class B reads 2,500/day free then $0.004/10,000; **Egress $0.00 via Cloudflare Bandwidth Alliance** (or free up to 3x stored volume directly) ([VERIFIED]).	B2 bucket set to **Private** (allPrivate) so nobody can guess raw backblazeb2.com URLs. Browser uploads via 15-min S3 presigned PUT URL. Public reads route via cdn.swiftradish.app Cloudflare Worker that signs origin GETs to B2 and sets Cache-Control: public, max-age=31536000, immutable.	Local **MinIO** container in Docker Compose using identical S3 aioboto3 calls. Switching from B2 to Cloudflare R2 or AWS S3 requires changing only env vars.
6. Realtime Broker & Background Jobs	**Redis 7 + arq (Asyncio Redis Queue)**. *Alternative evaluated:* FastAPI BackgroundTasks (loses jobs on container restart); Celery (heavy sync process overhead).	**Owns:** Ephemeral WebSocket Pub/Sub channels, arq job queues/retries, sliding-window rate limits, and JWT jti revocation denylist. **Does NOT Own:** Durable chat history or persistent state.	Upstash / Railway / Render Redis: 0–10/mo for 250MB–1GB instance (allkeys-lru disabled for queue DB; noeviction enforced).	Internal private network only (TLS + password). If Redis experiences a brief outage, API mutations still commit to Postgres and an outbox reconciliation sweep re-enqueues missed jobs.	Local redis:7-alpine in Docker Compose. Standard Redis protocol portable across any cloud provider.

7.3 Explicit Backblaze B2 + Cloudflare Configuration Verification Protocol

Because the user's Backblaze B2 bucket and Supabase project were created prior to this specification and have not been inspected ([UNVERIFIED]), five mandatory configuration steps must be executed and verified in Slice 0/1 before writing media upload code. First, the builder must inspect the B2 bucket visibility and region, confirming the bucket is set to **Private** (allPrivate) and recording its exact S3 endpoint (such as https://s3.us-west-004.backblazeb2.com). Second, explicit S3-compatible CORS rules must be applied on the B2 bucket permitting PUT, GET, and HEAD from http://localhost:3000 (development) and https://swiftradish.app (production), exposing the ETag header and allowing Content-Type and Content-Length headers (maxAgeSeconds: 3600). Third, two separate scoped Backblaze Application Keys must be generated: a read/write key for FastAPI and arq workers (s3:PutObject, s3:GetObject, s3:HeadObject, s3:DeleteObject scoped to the bucket), and a read-only Application Key (readFiles) restricted to the media/ prefix for the Cloudflare Delivery Worker. Fourth, Backblaze Bucket Info must be set to {"Cache-Control": "public, max-age=31536000, immutable"} alongside deploying the lightweight Cloudflare Worker (backblaze-b2-samples/cloudflare-b2 pattern) on cdn.swiftradish.app to sign GET/HEAD requests to the media/ prefix and override Cloudflare's default Authorization-header cache bypass, ensuring Cloudflare PoPs cache public images at the edge with **$0.00 Bandwidth Alliance egress**. Fifth, a B2 bucket lifecycle rule must be configured on staging/ to auto-cancel incomplete multipart uploads and purge objects in staging/ older than 24 hours as a storage-level safety net backing up the 6-hour arq orphan cleanup cron.

H. Domain/Data Model and API Contracts

8.1 Reconciled PostgreSQL 16 Domain Model

All tables reside in the public schema managed exclusively via **Alembic** migrations (Base.metadata.create_all() is strictly banned). Primary keys use **UUIDv7** (time-ordered 128-bit UUIDs) so all B-Tree index insertions are sequential and support deterministic keyset pagination (WHERE id < :cursor_id ORDER BY id DESC) without timestamp tie-breaker bugs.

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
users	**MVP Core (Slice 1)**	Core identity record synced 1:1 with Supabase auth.users.id. PK: id (UUID, matches Supabase sub).	username VARCHAR(24) NOT NULL, email VARCHAR(255) NOT NULL, birth_year SMALLINT NOT NULL CHECK	UNIQUE (LOWER(username)), UNIQUE (LOWER(email)), idx_users_username_trgm USING GIN (username	Soft-deleted for 14 days on user deletion request, then hard-purged by arq job. email and birth_year are never serialized to public DTOs.

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
			(birth_year <= EXTRACT(YEAR FROM NOW()) - 18), role VARCHAR(24) DEFAULT 'user' CHECK (role IN ('user','moderator','grievance_officer','admin')), status VARCHAR(24) DEFAULT 'ACTIVE' CHECK (status IN ('ACTIVE','SUSPENDED','PENDING_DELETION')), onboarding_step SMALLINT DEFAULT 1, consent_version VARCHAR(16) NOT NULL, consented_at TIMESTAMPTZ NOT NULL, deleted_at TIMESTAMPTZ.	gin_trgm_ops), idx_users_purge (status, deleted_at) WHERE status = 'PENDING_DELETION'.	
user_profiles	**MVP Core (Slice 1)**	Public/private persona and denormalized social counters. PK/FK: user_id -> users(id) ON DELETE	display_name VARCHAR(50) NOT NULL, bio VARCHAR(300) DEFAULT '', avatar_media[span_125](start_	idx_profiles_search_vec USING GIN (search_vector).	Readable by all unless user_blocks exists; if is_private = TRUE, non-followers

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
		CASCADE.	span)[span_125](end_span)_id UUID REFERENCES media_assets(id) ON DELETE SET NULL, theme_token VARCHAR(24) DEFAULT 'radish_crimson', is_private BOOLEAN DEFAULT FALSE, dm_policy VARCHAR(16) DEFAULT 'MUTUALS', follower_count INT DEFAULT 0 CHECK (follower_count >= 0), following_count INT DEFAULT 0 CHECK (following_count >= 0), search_vector TSVECTOR GENERATED ALWAYS AS (...) STORED.		only see header summary (display_name, avatar, bio) and cannot query posts or relationship lists.
user_relationships	**MVP Core (Slice 2)**	Directed social graph edge (actor_id follows target_id). PK: id (UUIDv7).	actor_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE, target_id UUID NOT NULL	idx_rel_actor_accepted (actor_id, target_id) WHERE status = 'ACCEPTED', idx_rel_target_status	PL/pgSQL trigger sets is_mutual = TRUE on both directional rows when reciprocal ACCEPTED rows exist and

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
			REFERENCES users(id) ON DELETE CASCADE, status VARCHAR(16) NOT NULL CHECK (status IN ('PENDING','ACCEPTED')), is_mutual BOOLEAN DEFAULT FALSE NOT NULL, UNIQUE (actor_id, target_id), CHECK (actor_id <> target_id).	(target_id, status, id DESC), idx_rel_mutual (actor_id, target_id) WHERE is_mutual = TRUE.	updates follower_count / following_count atomically.
user_blocks & user_mutes	**MVP Core (Slice 2/6)**	Bidirectional invisibility (user_blocks) and unidirectional feed suppression (user_mutes). Composite PK: (actor_id, target_id).	actor_id UUID REFERENCES users(id) ON DELETE CASCADE, target_id UUID REFERENCES users(id) ON DELETE CASCADE, created_at TIMESTAMPTZ DEFAULT NOW(), CHECK (actor_id <> target_id).	idx_blocks_reverse (target_id, actor_id).	Every feed, profile, search, Co-Plot, and comment query enforces NOT EXISTS against user_blocks in both directions.
patches & patch_members	**MVP Core (Slice 5)**	Topic communities and membership	slug VARCHAR(32) NOT NULL, name	UNIQUE (LOWER(slug)), idx_patches_tr	Public Patches readable by all; creating a post inside a Patch

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
		roles. PK: id (UUIDv7).	VARCHAR(60) NOT NULL, description VARCHAR(500) NOT NULL, rules JSONB DEFAULT '[]', creator_id UUID REFERENCES users(id) ON DELETE SET NULL, member_count INT DEFAULT 0. In patch_members: (patch_id, user_id) PK, role IN ('MEMBER','STEWARD').	gm USING GIN (name gin_trgm_ops, description gin_trgm_ops), idx_patch_members_user (user_id, patch_id).	requires an active patch_members row.
posts	**MVP Core (Slice 4)**	Standard posts and parent records for Co-Plots. PK: id (UUIDv7).	author_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE, patch_id UUID REFERENCES patches(id) ON DELETE SET NULL, post_type VARCHAR(16) NOT NULL CHECK (post_type IN ('STANDARD','CO_PLOT')), body TEXT NOT NULL	idx_posts_author_cursor (author_id, id DESC) WHERE deleted_at IS NULL, idx_posts_patch_cursor (patch_id, id DESC) WHERE deleted_at IS NULL, idx_posts_public_cursor (id DESC) WHERE visibility = 'PUBLIC' AND	Soft-delete sets deleted_at = NOW() and enqueues B2 object purge after 24h (unless moderation_status = 'REMOVED', which preserves encrypted evidence for 180 days per IT Rules Rule 3(1)(g)).

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
			CHECK (char_length(body) <= 2200), visibility VARCHAR(16) NOT NULL CHECK (visibility IN ('PUBLIC','FOLLOWERS','MUTUALS','PATCH_ONLY')), reaction_count INT DEFAULT 0, comment_count INT DEFAULT 0, moderation_status VARCHAR(24) DEFAULT 'ACTIVE' CHECK (moderation_status IN ('ACTIVE','PENDING_MOD_SHIELD','REMOVED')), search_vector TSVECTOR GENERATED ALWAYS AS (...) STORED, deleted_at TIMESTAMPTZ.	deleted_at IS NULL, idx_posts_fts USING GIN (search_vector).	
co_plots & co_plot_participants	**MVP Core (Slice 4)**	Structured coordination metadata and RSVP/submissi	title VARCHAR(120) NOT NULL, starts_at	idx_coplots_starts_at (starts_at ASC) WHERE starts_at IS	Only post.author_id can edit Co-Plot

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
		on tracking. PK/FK: post_id -> posts(id) ON DELETE CASCADE.	TIMESTAMPTZ , location_label VARCHAR(160), capacity INT CHECK (capacity IS NULL OR capac[span_13 4](start_span)[s pan_134](end_ span)ity > 0), going_count INT DEFAULT 1, contribution_po licy VARCHAR(24) DEFAULT 'MUTUALS_AU TO' CHECK (contribution_p olicy IN ('MUTUALS_A UTO','HOST_A PPROVAL','CL OSED')), conversation_id UUID REFERENCES conversations(i d). In co_plot_particip ants: UNIQUE (post_id, user_id), rsvp_status IN ('GOING','MAY BE','DECLINE D').	NOT NULL, idx_coplot_part _user (user_id, rsvp_status, post_id).	metadata or approve HOST_APPRO VAL carousel submissions.
media_assets	**MVP Core (Slice 3)**	Authoritative state machine for all uploaded	owner_id UUID NOT NULL REFERENCES	idx_media_post _order (post_id, position ASC)	Only owner_id can finalize or attach a

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
		images and collaborative carousel slides. PK: id (UUIDv7).	users(id) ON DELETE CASCADE, post_id UUID REFERENCES posts(id) ON DELETE CASCADE, contributor_id UUID REFERENCES users(id) ON DELETE SET NULL, b2_staging_key VARCHAR(255) UNIQUE, b2_public_prefix VARCHAR(255) UNIQUE, mime_type VARCHAR(32) NOT NULL, byte_size INT NOT NULL CHECK (byte_size <= 15728640), width INT, height INT, blurhash VARCHAR(64), alt_text VARCHAR(1000) DEFAULT '', position SMALLINT DEFAULT 0, status VARCHAR(24) NOT NULL	WHERE status = 'READY', idx_media_orphan_cleanup (status, expires_at) WHERE post_id IS NULL.	READY asset (post_id IS NULL) to a new post or Co-Plot submission. Unattached assets are purged after 6 hours.

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
			CHECK (status IN ('PENDING_UPLOAD','VERIFYING','READY','PENDING_APPROVAL','REJECTED','FAILED','QUARANTINED')), expires_at TIMESTAMPTZ NOT NULL.		
post_reactions, comments & bookmarks	**MVP Core (Slice 5)**	Reactions, 3-level nested comments, and private saves. PK: id (UUIDv7).	post_reactions: UNIQUE (post_id, user_id, reaction_code). comments: parent_id UUID REFERENCES comments(id), depth SMALLINT NOT NULL CHECK (depth BETWEEN 0 AND 2), is_tombstone BOOLEAN DEFAULT FALSE. bookmarks: UNIQUE (user_id, post_id).	idx_comments_post_tree (post_id, parent_id, id ASC), idx_bookmarks_user_cursor (user_id, id DESC).	Deleting a comment that has child replies sets body = NULL, is_tombstone = TRUE so nested child replies remain readable. Bookmarks are strictly private to user_id.
conversations, conversation_participants & messages	**MVP Core (Co-Plot Thread in Slice 4); Expanded to 1:1/Group**	Durable chat threads for Co-Plots (MVP Core) and 1:1/Group DMs (Stage 2). PK:	conversations: type IN ('CO_PLOT_THREAD','DIRECT','GROUP'), direct_pair_key	idx_messages_conv_cursor (conversation_id, id DESC), idx_conv_part_user_sort	Reading/sending messages requires an ACTIVE row in conversation_participants and

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
	DMs in Stage 2 (Slice 7)	id (UUIDv7).	VARCHAR(80) UNIQUE, last_message_id UUID, last_message_at TIMESTAMPTZ. conversation_participants: UNIQUE (conversation_id, user_id), last_read_message_id UUID, state IN ('ACTIVE','REQUEST_PENDING','LEFT'). messages: client_msg_id UUID NOT NULL, UNIQUE (conversation_id, sender_id, client_msg_id).	(user_id, last_message_at DESC).	zero block edge.
notifications, safety_reports & audit_events	**MVP Core (Slice 6)**	Deduplicated activity center, statutory SLA tickets, and immutable moderation ledger. PK: id (UUIDv7).	notifications: UNIQUE (recipient_id, dedupe_key), actor_count INT DEFAULT 1, read_at TIMESTAMPTZ. safety_reports: ticket_number VARCHAR(24) UNIQUE, sla_tier IN ('CRITICAL_24H','STANDARD	idx_notif_recipient_cursor (recipient_id, id DESC), idx_reports_sla_queue (status, sla_deadline_at ASC).	PostgreSQL trigger on audit_events raises an exception on any UPDATE or DELETE to guarantee 180-day statutory evidentiary integrity.

Table Name	Scope Stage	Purpose, PK & Foreign Keys	Key Columns, Constraints & Lifecycle States	Critical Indexes (B-Tree, GIN, Partial)	Retention, Deletion & Server Authorization Rules
			_72H','LEGAL_72H','GENERAL_15D'), sla_deadline_at TIMESTAMPTZ NOT NULL, status IN ('OPEN','UNDER_REVIEW','ACTION_TAKEN','DISMISSED','APPEALED'). audit_events: append-only JSONB evidence log.		

8.2 Canonical API Contracts by User Action

All endpoints are prefixed with /api/v1, authenticate via Authorization: Bearer <supabase_es256_jwt> (or HttpOnly cookie forwarded by Next.js BFF), validate inputs via **Pydantic v2 (extra='forbid')**, enforce two-plane server-side authorization (Plane 1 Role + Plane 2 Object Visibility & Block Check), and return **RFC 9457 Problem Details JSON** ({"type", "title", "status", "detail", "instance", "error_code", "request_id"}) on all 4xx/5xx responses. Private resources always return 404 Not Found (never 403 Forbidden) to unauthorized callers to prevent ID enumeration.

User Action	Method & Path	Auth & Rate Limit (Redis Sliding Window)	Request Shape & Validation Rules	Sync vs. Async Execution & Side Effects	Durable Success Criteria & Response (2xx / Error Codes)
1. Register 18+ Account	POST /api/v1/auth/register	anon | 5 req / 15m per IP	{"email": EmailStr, "password": str (10..128 chars), "birth_date": date, "consent_accepted": Literal[True], "consent_version": "2026.1"}.	**Sync:** Validates age >= 18, creates/syncs Supabase Auth user, triggers OTP email. **Async:** None.	202 Accepted {"status": "AWAITING_OTP"}. Errors: 422 ERR_AGE_RESTRICTED (<18), 429 ERR_RATE_LIMITED.
2. Request	POST	auth | 30 files /	{"files":	**Sync:** Inserts	201 Created

User Action	Method & Path	Auth & Rate Limit (Redis Sliding Window)	Request Shape & Validation Rules	Sync vs. Async Execution & Side Effects	Durable Success Criteria & Response (2xx / Error Codes)
Media Upload Intents	/api/v1/media/upload-intents	1h per user	[{"client_ref": str, "filename": str, "content_type": Literal["image/jpeg","image/png","image/webp","image/[span_121](start_span)[span_121](end_span)avif"], "byte_size": int (1..15728640)}] } (max 10 files).	media_assets rows (status='PENDING_UPLOAD', expires_at=NOW()+1h), generates 15-min B2 S3 presigned PUT URLs (staging/{user_id}/{media_id}) with signed Content-Type and Content-Length.	returning array of {"media_id": UUID, "upload_url": str, "required_headers": {...}, "expires_at": datetime}. Errors: 413 ERR_FILE_TOO_LARGE, 415 ERR_UNSUPPORTED_MEDIA, 429.
3. Finalize & Verify Uploaded Media	POST /api/v1/media/{media_id}/complete	auth (Owner only) | 60 req / 1h	{"alt_text": str (0..1000 chars), "crop_box": Optional[CropRect]}.	**Sync (Fast Path \le 600\text{ms} via asyncio.to_thread):** Executes B2 S3 HeadObject + Range: bytes=0-4095 signature verification, runs pyvips streaming decode (max 8192x8192px bomb check), strips all EXIF/GPS tags, generates Blurhash, writes thumb_400w.webp, feed_1080w.we	200 OK {"media_id": UUID, "status": "READY", "width": int, "height": int, "blurhash": str, "urls": {"thumb": str, "feed": str, "full": str}}. UI may enable Plant Post only when all attached assets return READY. Errors: 400 ERR_UPLOAD_INCOMPLETE, 422 ERR_MALFORMED_IMAGE.

User Action	Method & Path	Auth & Rate Limit (Redis Sliding Window)	Request Shape & Validation Rules	Sync vs. Async Execution & Side Effects	Durable Success Criteria & Response (2xx / Error Codes)
				bp, full_2048w.webp to media/{media_id}/, deletes staging/ object, and updates status='READY'.	
4. Publish Post or Co-Plot	POST /api/v1/posts	auth | 15 posts / 1h per user; requires Idempotency-Key header	{"body": str (1..2200), "visibility": VisibilityEnum, "patch_id": Optional[UUID], "media_ids": list[UUID] (0..10), "co_plot": Optional[{"title": str, "starts_at": Optional[datetime], "location_label": Optional[str], "capacity": Optional[int], "contribution_policy": PolicyEnum}]}.	**Sync:** Verifies Idempotency-Key in Redis (24h TTL). Verifies all media_ids belong to current_user.id, have status='READY', and post_id IS NULL. Atomically inserts posts, optional co_plots + conversations, and links media_assets.	201 Created with full PostDetailDTO. UI wipes composer state, revokes blob URLs, and deletes IndexedDB draft only after 201. Errors: 403 ERR_NOT_PATCH_MEMBER, 409 ERR_MEDIA_ALREADY_ATTACHED, 422.
5. Fetch Home Feeds (My Patch & Fresh Harvest)	GET /api/v1/feed/following & GET /api/v1/feed/discover	auth | 120 req / 1m per user	Query params: ?cursor=<uuidv7_or_score_cursor>&limit=20 (limit bounded 1..30).	**Sync:** Executes single keyset query with selectinload() for author profile, media assets, Co-Plot summary, and viewer's reaction/RSVP state (strictly 0	200 OK {"items": list[PostCardDTO], "next_cursor": Optional[str], "caught_up_post_id": Optional[UUID]}.

User Action	Method & Path	Auth & Rate Limit (Redis Sliding Window)	Request Shape & Validation Rules	Sync vs. Async Execution & Side Effects	Durable Success Criteria & Response (2xx / Error Codes)
				N+1 queries; excludes user_blocks and user_mutes).	
6. RSVP to Co-Plot	POST /api/v1/plots/{post_id}/rsvp	auth | 30 req / 15m per user	{"rsvp_status": Literal["GOING", "MAYBE", "DECLINED"]}.	**Sync:** Checks visibility & block rules, locks co_plots row (FOR UPDATE), checks capacity, upserts co_plot_participants, updates going_count, grants conversation_participants access if GOING. **Async (arq):** Dispatches deduplicated notification to host.	200 OK {"rsvp_status": str, "going_count": int, "conversation_id": Optional[UUID]}. Errors: 404 (private/blocked), 409 ERR_PLOT_FULL.
7. Submit Collaborative Slide to Co-Plot	POST /api/v1/plots/{post_id}/submissions	auth | 20 slides / 1h per user	{"media_ids": list[UUID] (1..3)}. Requires caller to have READY unattached media.	**Sync:** Checks co_plots.contribution_policy. If MUTUALS_AUTO and caller is_mutual == TRUE with host, attaches slides directly (status='READY', post_id=post_id, contributor_id=user.id).	201 Created {"submission_status": "APPROVED" | "PENDING_APPROVAL", "attached_media_ids": list[UUID]}.

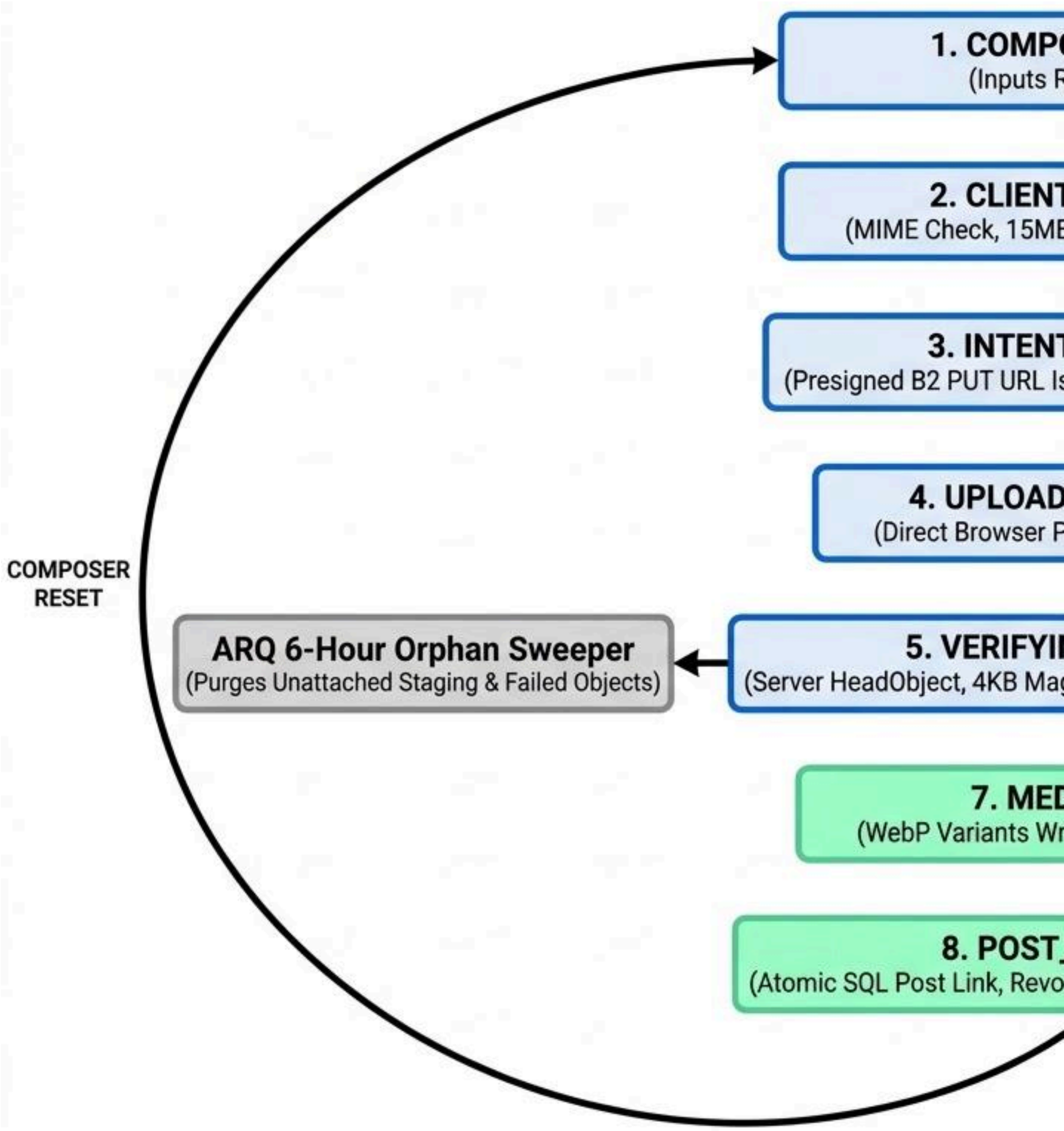
User Action	Method & Path	Auth & Rate Limit (Redis Sliding Window)	Request Shape & Validation Rules	Sync vs. Async Execution & Side Effects	Durable Success Criteria & Response (2xx / Error Codes)
				Otherwise sets status='PENDING_APPROVAL'. **Async (arq):** Notifies host.	
8. Unified Hybrid Search	GET /api/v1/search	auth | 60 req / 1m per user	Query params: ?q=<str (1..100 chars)>&type=all|users|posts|patches&cursor=<str>&limit=20.	**Sync:** Executes weighted PostgreSQL tsvector (websearch_to_tsquery) + pg_trgm similarity query (statement_timeout=2000ms), filtering out blocked users and private non-followed content.	200 OK {"users": [...], "patches": [...], "posts": [...], "did_you_mean": Optional[str], "next_cursor": Optional[str]}.
9. Submit Statutory Safety Report	POST /api/v1/safety/reports	auth (or rate-limited anon on /legal/grievance) | 10 req / 1h	{"target_type": Literal["POST","COMMENT","USER","MESSAGE"], "target_id": UUID, "category": ReportCategoryEnum, "details": str (0..1000),[span_127](start_span)[span_127](end_span) "block_target_user": bool}.	**Sync:** Assigns ticket_number and sla_tier (CRITICAL_24H, STANDARD_72H, GENERAL_15D), writes safety_reports and audit_events, executes atomic block if block_target_user=True. **Async (arq <60\text{s}):** Sends statutory acknowledgme	201 Created {"ticket_number": str, "sla_tier": str, "sla_deadline_at": datetime, "acknowledged_at": datetime}.

User Action	Method & Path	Auth & Rate Limit (Redis Sliding Window)	Request Shape & Validation Rules	Sync vs. Async Execution & Side Effects	Durable Success Criteria & Response (2xx / Error Codes)
				nt email & notification.	

I. Media, Messaging, Notification, and Job Lifecycle Specifications

9.1 Direct-to-Backblaze B2 Media Pipeline and Composer Reset State Machine

A successful browser upload to Backblaze B2 (HTTP 200 OK on presigned PUT) **never** equates to a published post or a verified media asset. The system enforces an eight-state lifecycle across the client and server so malformed files, decompression bombs, and network drops are isolated before any post is committed.



Authoritative Server Validation & Security Controls (State 5: VERIFYING_PROCESSING)

Upon receiving POST /api/v1/media/{id}/complete, FastAPI executes an S3 HeadObject call

against staging/{user_id}/{media_id} in Backblaze B2 to verify that the object exists and its actual ContentLength is \le 15,728,640\text{ bytes} (15\text{MB}), then fetches Range: bytes=0-4095 to verify the binary magic bytes (FF D8 FF for JPEG, 89 50 4E 47 0D 0A 1A 0A for PNG, 52 49 46 46 .... 57 45 42 50 for WebP, or ftypavif for AVIF). SVG uploads by end-users are completely banned to eliminate stored XSS polyglots. Inside asyncio.to_thread(), pyvips inspects image header dimensions *before* allocating pixel buffers and immediately aborts (422 ERR_IMAGE_DIMENSIONS_EXCEEDED) if width > 8192, height > 8192, or width * height > 64,000,000 (64 megapixels). Valid images are decoded in streaming chunks (< 30\text{MB} RAM), stripped of all EXIF, IPTC, XMP, and GPS location tags (strip=True), converted to sRGB, and re-encoded into fresh WebP containers (thumb_400w.webp at quality 78, feed_1080w.webp at quality 82, and full_2048w.webp at quality 85), completely neutralizing any trailing polyglot payload appended to the original file. Once the three WebP variants are written to media/{media_id}/, the raw object in staging/{user_id}/{media_id} is immediately deleted via DeleteObject, and media_assets.status transitions to READY. Any media_assets row that remains in PENDING_UPLOAD, FAILED, or READY with post_id IS NULL past its 6-hour window is swept hourly by the arq purge_orphan_media cron job, which deletes both the B2 objects and the database row.

9.2 Realtime Co-Plot Thread & Messaging Lifecycle (MVP Core + Stage 2 Expansion)

Swiftradish guarantees **Durable Persistence Before Server Acknowledgement**, **At-Least-Once Delivery with Idempotent Deduplication**, and **Per-Conversation Sequential Ordering (UUIDv7)**, while explicitly avoiding unachievable claims of network-level "exactly-once" transport or guaranteed mobile background wake-up. To authenticate WebSocket connections without leaking JWTs in URL query strings or proxy access logs, the client requests a 30-second single-use ticket via POST /api/v1/ws/ticket (which writes SETEX ws_ticket:{nonce} 30 {user_id} in Redis) and connects to wss://api.swiftradish.app/ws?ticket={nonce} from the root authenticated layout (src/app/(main)/layout.tsx). FastAPI atomically consumes the ticket (GETDEL ws_ticket:{nonce}) and enforces a maximum of **5 concurrent WebSocket connections per user**, closing the oldest socket with code 4008 if exceeded. Every room subscription frame ({"type": "room.subscribe", "conversation_id": "..."}) and message frame ({"type": "msg.send", ...}) re-verifies in PostgreSQL and Redis that current_user.id holds an ACTIVE row in conversation_participants and is not blocked by the host or counterparty. When a user leaves a Co-Plot, is removed by the host, or is blocked, the server immediately publishes an internal control event (ws.revoke_subscription) to Redis Pub/Sub that detaches their socket from that conversation room in real time.

When sending a message, the client generates a client_msg_id (UUIDv4), renders an optimistic bubble (status = 'sending'), and transmits a msg.send frame. The server executes INSERT INTO messages (id, conversation_id, sender_id, client_msg_id, body) VALUES (:uuidv7, :conv_id, :user_id, :client_msg_id, :body) ON CONFLICT (conversation_id, sender_id, client_msg_id) DO UPDATE SET body = EXCLUDED.body RETURNING id, created_at. Only *after* the PostgreSQL transaction commits does the server return {"type": "msg.ack", "client_msg_id": "...", "server_msg_id": "<uuidv7>", "created_at": "..."} to the sender and publish msg.new to the Redis channel conv:{conversation_id}. Client and server exchange ping/pong heartbeats every 25s. On network disconnect, the client queues outgoing messages in IndexedDB (pendingOutbox), reconnects using exponential backoff (1s, 2s, 4s, 8s plus jitter),

sends {"type": "sync.resume", "conversation_id": "...", "after_msg_id": "<last_seen_uuidv7>"} to fetch all missed durable messages in strict id ASC order, and then flushes pendingOutbox, which deduplicates safely via client_msg_id.

9.3 Notification Deduplication and Background Job (arq) Lifecycle

Under the notification deduplication contract, ambient events (reactions, new followers, Co-Plot joins) compute a deterministic dedupe_key (react:{post_id}:{yyyy-mm-dd} or plot_join:{post_id}:{yyyy-mm-dd}) and execute INSERT INTO notifications ... ON CONFLICT (recipient_id, dedupe_key) DO UPDATE SET actor_count = notifications.actor_count + 1, last_actor_id = EXCLUDED.last_actor_id, updated_at = NOW(), read_at = NULL, preventing notification floods while bubbling the updated rollup row to the top of SCR-31. Every background arq task (process_orphan_media_sweep, dispatch_notification_fanout, send_statutory_ack_email, execute_account_export, execute_account_hard_purge) is architected to be **idempotent** (safe to run multiple times with the same job ID) and enforces a pessimistic execution timeout (60s for standard jobs, 300s for account export and purge). Failed jobs retry up to max_tries = 4 with exponential backoff (10s, 60s, 300s, 1800s); if all four attempts fail, the on_job_failure hook writes the payload, traceback, and retry count into a durable PostgreSQL dead_letter_jobs table and fires a Sentry alert for manual inspection and 1-command replay.

J. Security, Privacy, Moderation, Accessibility, and Legal-Review Register

10.1 Comprehensive Threat Model and Mitigation Matrix

Threat ID & Vector	Attacker Goal & Abuse Mechanism	Concrete Architectural Mitigation	Mandatory Automated Verification Test
SEC-01: Broken Object-Level Authorization (BOLA / IDOR)	Read private posts/Co-Plots, approve someone else's Co-Plot submissions, or attach another user's uploaded media_id to a new post.	Every service method enforces **Two-Plane Authorization**: Plane 1 checks RBAC role; Plane 2 queries object ownership (owner_id == user.id), visibility (PUBLIC, FOLLOWERS, MUTUALS, PATCH_ONLY), and bidirectional NOT EXISTS (user_blocks). Unauthorized calls return 404 Not Found.	pytest BOLA matrix executing cross-user read, update, delete, RSVP, and media-attach calls across all endpoints.
SEC-02: JWT Algorithm Confusion & JWKS SSRF	Forge tokens using Supabase's public key as an HMAC secret	Pin EXPECTED_ISSUER and	Unit & integration tests attempting alg: none, alg: HS256 signed with

Threat ID & Vector	Attacker Goal & Abuse Mechanism	Concrete Architectural Mitigation	Mandatory Automated Verification Test
	(CVE-2026-102266) or pass a malicious iss claim to trigger SSRF via PyJWKClient.	SUPABASE_JWKS_URL strictly from server env vars (never from unverified token payload). Enforce algorithms=["ES256"] (never including HS256 in the same decoder), require audience="authenticated", and check Redis revoked_jti:{jti} on mutations.	public key, expired tokens, wrong iss/aud, and revoked jti.
SEC-03: Malicious File Uploads, Polyglots & Decompression Bombs	Upload an HTML/SVG script disguised as .jpg, leak home GPS coordinates via EXIF, or crash FastAPI RAM with a 100,000x100,000px PNG pixel flood.	Ban SVG uploads; enforce 15MB presigned PUT limit; inspect first 4KB magic bytes on server; enforce 8192x8192px (64MP) header limit in pyvips prior to decode; strip all EXIF/GPS tags (strip=True); re-encode every pixel buffer into a clean WebP container; serve from cdn.swiftradish.app with Content-Security-Policy: default-src 'none' and X-Content-Type-Options: nosniff.	Upload test suite with EICAR test file, SVG-in-JPEG polyglot, GPS-tagged photo (asserting zero GPS tags in output), and 100MP header bomb.
SEC-04: Private Bucket Enumeration & CDN Cache Poisoning	Guess Backblaze B2 URLs for deleted posts, private DM attachments, or account exports.	B2 bucket set to **Private** (allPrivate); object keys use unguessable UUIDv7 paths; public media served via Cloudflare Worker scoped strictly to media/ prefix; deleted posts immediately trigger DeleteObject on B2 and Cloudflare cache purge API.	Integration test verifying direct B2 URL returns 401/403 and deleted post CDN URL returns 404 after purge job.

Threat ID & Vector	Attacker Goal & Abuse Mechanism	Concrete Architectural Mitigation	Mandatory Automated Verification Test
SEC-05: Stored XSS, CSRF & Open Redirects	Inject javascript: links in Markdown posts/bios, forge cross-site mutations, or hijack /login?next= redirects.	Ban dangerouslySetInnerHTML via ESLint; render Markdown AST to React nodes allowing only [span_155](start_span)[span_155](end_span)https:// and mailto: links; enforce SameSite=Lax cookies + bearer header/origin check; validate next redirect param starts with single /.	OWASP XSS payload dictionary fuzz test across posts, bios, comments, and chat; open-redirect test suite.
SEC-06: Race Conditions in Capacity, Reactions, Follows & Blocks	Double-tap reactions/RSVPs to corrupt counters, exceed Co-Plot capacity, or message a user concurrently while being blocked.	Composite UNIQUE constraints + ON CONFLICT DO NOTHING on reactions, relationships, bookmarks, and client_msg_id; SELECT ... FOR UPDATE row lock on Co-Plot capacity checks; atomic block transaction that revokes active WebSocket subscriptions.	Concurrent asyncio.gather() test firing 20 parallel RSVP and reaction requests against a capacity-1 Co-Plot.
SEC-07: Spend Attacks & Resource Exhaustion	Flood presigned upload intents, search queries, or OTP requests to inflate cloud bills.	Redis sliding-window rate limits per user/IP (30 uploads/hr, 15 posts/hr, 60 searches/min, 5 auth attempts/15m), statement_timeout=5000ms, and a global Redis kill-switch (UPLOADS_ENABLED=false).	Rate-limit integration test asserting HTTP 429 + Retry-After header on the (N+1)\text{th} burst request.

10.2 Legal Accuracy Gate: India Regulatory Review (DPDP Act 2023, DPDP Rules 2025 & IT Rules 2021)

The following analysis synthesizes verified primary Indian statutory texts and official MeitY notifications to guide technical architecture and workflow design; it does not constitute formal

legal advice, and specific operational items are flagged for review by qualified Indian legal counsel ([VERIFIED], High Confidence).

Statutory Framework & Provision	Primary Legal Source & Current Enforcement Status (Oct 2026)	Technical & Product Controls Specified in Swiftradish	Items Flagged for Qualified Indian Legal Counsel
DPDP Act, 2023 & DPDP Rules, 2025 — Commencements & Timelines	Enacted Aug 11, 2023; **DPDP Rules, 2025 notified by MeitY on Nov 13–14, 2025** (G.S.R. 843(E), 844(E), 846(E)). Data Protection Board (DPB) established **Nov 14, 2025**; Consent Managers take effect **Nov 14, 2026**; full substantive compliance takes effect **May 13–14, 2027** (with a Jan 23, 2026 MeitY consultation proposing acceleration to **Nov 2026**). Until May 2027, the IT Act 2000 & SPDI Rules continue to apply in parallel ([VERIFIED]).	Architect all consent, security, erasure, and breach-reporting workflows to meet the DPDP Rules 2025 standard immediately at launch rather than waiting for May 2027.	Confirm whether MeitY issues any formal gazette notification accelerating substantive enforcement ahead of May 14, 2027.
DPDP Section 5 & 6 — Notice & Consent (No "Legitimate Interest")	Unlike GDPR, the DPDP Act **does not recognize "legitimate interest"**. Consent must be free, specific, informed, unconditional, and unambiguous (Sec 6(1)), preceded by a standalone itemized notice detailing the personal data collected, specific purpose, right to withdraw consent, and grievance mechanism, accessible across the **22 Eighth Schedule languages** ([VERIFIED]).	SCR-04 (/register) presents a standalone, unbundled consent notice recording consent_version and consented_at in users, plus a 1-click consent withdrawal / account deletion flow in SCR-33/37.	Review the exact English and 22 Eighth Schedule language translations of the Privacy Notice and Terms of Service.
DPDP Section 9 &	Processing personal	Swiftradish enforces an	**Counsel Required:**

Statutory Framework & Provision	Primary Legal Source & Current Enforcement Status (Oct 2026)	Technical & Product Controls Specified in Swiftradish	Items Flagged for Qualified Indian Legal Counsel
DPDP Rules — Protection of Children's Data (<18)	data of any individual under 18 requires verifiable parental consent (via DigiLocker or virtual tokens issued by an authorized entity) and strictly prohibits behavioral monitoring or targeted advertising to children (maximum penalty up to **₹200 crore**) ([VERIFIED]).	explicit **18+ age gate** at registration (SCR-04) and prohibits behavioral ad tracking across the entire platform.	Verify whether self-declared DOB at registration suffices when the platform explicitly prohibits under-18 users, or when DigiLocker age-token verification becomes legally required for adult-only platforms under the notified DPDP Rules.
DPDP Section 8(5), Section 8(6) & Rule 7 — Security Safeguards & 72-Hour Breach Notification	Sec 8(5) mandates reasonable security safeguards including encryption in transit/at rest, access controls, and logs (penalty up to **₹250 crore**). Sec 8(6) and **Rule 7** mandate notifying the DPB and affected Data Principals within **72 hours** of discovering a personal data breach, with **no materiality threshold** (penalty up to **₹200 crore**) ([VERIFIED]).	Enforce TLS 1.3, encrypted Supabase Postgres & B2 storage, Two-Plane RBAC/BOLA checks, and an administrative **Breach Notification Runbook & Dispatch Tool** capable of emailing affected users and generating the statutory DPB report within 24–72 hours.	Approve the legal text of the statutory DPB breach notification template and Data Processor contractual clauses (Sec 8(2)).
IT Rules, 2021 (as amended Oct 2022 & Apr 2023) — Rule 3 Grievance SLAs & GAC Appeals	**Rule 3(2)(a)(i):** Grievance Officer must acknowledge complaints within **24 hours**, resolve general complaints within **15 days**, and resolve complaints requesting removal of prohibited content under **Rule 3(1)(b)** (except sub-clauses (i), (iv), (xi)) within **72 hours**. **Rule 3(2)(b):** Content violating bodily privacy, non-consensual	Publish Grievance Officer name and contact details on /legal/grievance and SCR-38; auto-acknowledge all reports in <60\text{s} with a unique ticket number; enforce 24h, 72h, and 15d queues in SCR-39; preserve removed evidence in append-only audit_events + B2 quarantine for 180 days; surface dispute	Designate the named Grievance Officer prior to public launch; monitor Indian registered user count against the **50 lakh (5 million) SSMI threshold** (Rule 4) and the **2 crore (20 million) DPDP Third Schedule retention threshold**.

Statutory Framework & Provision	Primary Legal Source & Current Enforcement Status (Oct 2026)	Technical & Product Controls Specified in Swiftradish	Items Flagged for Qualified Indian Legal Counsel
	imagery, or impersonation/morphing must be removed within **24 hours**. **Rule 3(1)(g):** Preserve removed records for **180 days**. **Rule 3(1)(j):** Law enforcement orders within **72 hours**. **Rule 3A:** Users may appeal to the **Grievance Appellate Committee (gac.gov.in)** within **30 days**.	and GAC appeal links in SCR-38.	

K. Reliability, Observability, Test Strategy, and Measurable SLOs

11.1 Measurable Service-Level Objectives (SLOs) and Release Gates

All performance and reliability targets are formulated as measurable engineering gates verified in CI/CD prior to production release ([RECOMMENDATION], High Confidence).

Subsystem / Metric	Proposed Target Budget	Measurement Method & Tooling	Automated Release Gate Test	Degraded Mode / Fallback Behavior
Frontend Initial Load & Core Web Vitals	LCP <= 1.8s, INP <= 120ms, CLS <= 0.03, Initial JS bundle < 140KB gzipped (Mid-tier mobile 4G throttle).	Lighthouse CI on /, /home, and /p/[postId] + @next/bundle-analyzer in GitHub Actions.	CI fails if initial route JS exceeds 140KB gzipped or CLS > 0.03.	Server-stored width/height + Blurhash canvas prevents layout shift even if CDN image delivery is slow.
REST API Latency & SQL Query Budget	Read endpoints (GET /feed/*, GET /posts/{id}): p95 <= 80ms, p99 <= 180ms. Write endpoints (POST /posts, POST /rsvp): p95 <= 150ms. Individual SQL queries: p95 <= 15ms.	OpenTelemetry / Structlog middleware recording duration_ms and SQL query count per request.	pytest SQL query-counter assertion (assert query_count <= 3 on feed fetch) + k6 load test (100 VUs for 3m).	PgBouncer enforces statement_timeout = 5000ms; Discover feed falls back to chronological idx_posts_public_cursor if ranking query latency exceeds 150ms.
Media	Presigned intent	Timer histogram	Integration test	If pyvips thread

Subsystem / Metric	Proposed Target Budget	Measurement Method & Tooling	Automated Release Gate Test	Degraded Mode / Fallback Behavior
Verification & Variant Generation	generation: p95 <= 60ms. POST /media/{id}/complete (HeadObject + pyvips 3 WebP variants): p95 <= 650ms.	around pyvips worker thread and B2 S3 calls.	uploading 5 concurrent 4MB JPEGs and asserting p95 < 1000ms and < 150MB worker RSS memory delta.	pool is saturated, complete returns 202 Accepted (status='VERIFYING') and offloads resize to arq worker while client polls GET /media/{id}/status every 1.5s.
Realtime Co-Plot Thread & WebSocket Fanout	Socket message fanout (msg.send -> recipient msg.new): p95 <= 120ms. Reconnect gap sync (sync.resume): p95 <= 150ms.	k6 WebSocket test script measuring round-trip msg.ack and cross-client msg.new timestamps.	Playwright network-drop test verifying zero lost or duplicated messages across a 10-second offline toggle.	If WebSocket upgrade is blocked by a corporate proxy or container restart, client automatically falls back to POST /messages + 15s REST cursor polling.
Background Queue (arq) & Statutory Acknowledgment	Statutory report acknowledgment (safety_reports -> ack email/notification): p99 <= 60s (vs 24h legal limit). Queue lag: < 30s.	Redis queue depth gauge (LLEN arq:queue) and job duration telemetry.	Integration test submitting a report and asserting ack notification row and email dispatch within 5s.	Synchronous database transaction generates ticket_number and returns it immediately in the HTTP 201 response (SCR-36) even if the email worker is delayed.

11.2 Observability, Privacy-Safe Logging, Backups, and PWA Platform Truths

All FastAPI and arq logs emit JSON via structlog including request_id, user_id (UUID only), route, status_code, and duration_ms. A mandatory scrubbing processor strips Authorization headers, cookies, passwords, OTPs, raw email addresses, IP addresses (which are hashed with a rotating daily salt for rate-limit forensics), post bodies, and message contents so private user content never enters log aggregation systems such as Sentry or Grafana Cloud. For database integrity and disaster recovery, FastAPI's lifespan startup hook executes SELECT version_num FROM alembic_version and compares it against the codebase's expected HEAD revision, halting boot immediately on any mismatch. Nightly automated logical backups (pg_dump --format=custom encrypted via age or GPG) are uploaded to an isolated backups/ prefix in Backblaze B2 with **Object Lock** enabled, and a monthly CI/staging job restores the

latest backup into an ephemeral Postgres 16 container and runs alembic current plus row-count integrity checks (RTO < 30 minutes, RPO < 24 hours on logical backup or < 5 minutes with Supabase Pro PITR).
Regarding browser and iOS PWA capabilities ([VERIFIED]), on **iOS and iPadOS (16.4 through iOS 19)**, Web Push notifications **only** function when the user manually adds Swiftradish to their Home Screen via Safari's Share sheet (display: "standalone") and subsequently taps an explicit in-app button that triggers Notification.requestPermission(), as standard mobile Safari tabs cannot receive Web Push. Silent background pushes are also forbidden on iOS, meaning the Service Worker (sw.js) must invoke self.registration.showNotification() on every push event. Consequently, the **in-app Activity Center (SCR-31) serves as the primary reliable notification channel in MVP Core**, while Web Push (SCR-10) is presented strictly as an opt-in enhancement for installed PWA users without promising guaranteed background delivery.

L. Current Cost Model and Resource Realism

12.1 Official 2026 Pricing Inputs and Mathematical Cost Formulas

The financial model is recalculated from verified 2026 official vendor pricing ([VERIFIED], High Confidence). For **Backblaze B2 Object Storage**, the first 10 GB of storage is free, followed by **$0.006 / GB / month** $(6.00/TB/mo)$; $ClassAwrites(uploads)arefree(0.00)$; Class B reads (GetObject, HeadObject) include 2,500 / day free, followed by **$0.004 per 10,000 requests** ($0.40 per million); and **egress via the Cloudflare Bandwidth Alliance is $0.00 / GB** (while direct B2 egress is free up to 3\times average monthly stored volume, then $0.01/GB). By comparison, **Cloudflare R2** offers 10 GB free storage and then charges **$0.015 / GB / month** ($15.00/TB/mo), with 1M/mo free Class A writes (then **$4.50 / million**), 10M/mo free Class B reads (then **$0.36 / million**), and **$0.00 / GB** egress. For **Supabase (Database + Auth)**, the Free Tier ($0/mo: 500MB DB, 50,000 MAU Auth, pausing after 7 days of inactivity) is suitable only for initial development, whereas **Supabase Pro ($25/mo fixed base)** includes 8 GB database storage (then **$0.125 / GB / mo**) and 100,000 MAU Auth (then **$0.00325 / MAU**).
\text{Monthly
M[span_159](start_span)[span_159](end_span)[span_161](start_span)[span_161](end_span)[span_163](start_span)[span_163](end_span)edia Storage Cost (B2 + CF CDN)} = \max(0, V_{\text{GB}} - 10) \times \$0.006 + \max(0, R_{\text{uncached}} - 75,000) \times \frac{\$0.004}{10,000} + \$0.00\text{ (Bandwidth Alliance Egress)} \text{Monthly Media Volume } \Delta V_{\text{GB}} = \text{Posts}_{\text{mo}} \times \text{AvgImages}_{\text{post}} \times \text{AvgVariantsSize}_{\text{MB}} \div 1024
Assuming an average post or Co-Plot attaches 3.0 images, and pyvips generates three optimized WebP variants (thumb_400w \approx 35\text{KB}, feed_1080w \approx 160\text{KB}, full_2048w \approx 380\text{KB}; total **0.575\text{MB} per image** across all 3 variants after deleting the raw staging upload), each 3-image post consumes **1.725\text{MB}** of permanent object storage.

12.2 Low, Base, and High Scenario Cost Breakdown Table

Cost Driver / Infrastructure Layer	Low Scenario (Seed Beta: 1,000 MAU / 250 DAU, 1,500 posts/mo)	Base Scenario (Growing Community: 10,000 MAU / 3,000 DAU, 20,000 posts/mo)	High Scenario (Scaled Traction: 50,000 MAU / 15,000 DAU, 120,000 posts/mo)	Dominant Variable & Cost Guardrails
1. Supabase PostgreSQL 16 + Supabase Auth	**$0/mo** (Dev/Closed Alpha on Free Tier) or **$25.00/mo** (Production Launch on Supabase Pro: ~1.5 GB DB, 1k MAU).	**$26.50/mo** (Supabase Pro $25 base + 12 GB total DB = 4 GB overage \times \$0.125; 10k MAU included in 100k cap).	**$65.25/mo** ($25 Pro base + ~50 GB DB storage = $5.25 + Small Compute Add-on $35/mo for higher connection/CPU headroom; 50k MAU included).	**Dominant variable:** Database row & GIN index size (posts, messages, notifications). Guardrail: Prune read notifications older than 90 days.
2. FastAPI + arq Worker + Next.js Compute	**$14.00/mo** (1x 1GB FastAPI+WS container $7 + 1x 512MB arq worker $5 on Railway/Fly.io; Next.js on Vercel Hobby $0 or container $2).	**$64.00/mo** (2x 2GB FastAPI+WS containers $28 + 2x 1GB arq workers $16 + Next.js Pro/Container $20).	**$210.00/mo** (5x 2-vCPU/4GB FastAPI+WS nodes $125 + 3x 2GB arq workers $45 + 2x Next.js SSR containers $40).	**Dominant variable:** Concurrent WebSocket connections (~8,000 per 4GB node) and pyvips CPU bursts. Guardrail: 30 uploads/hr cap.
3. Backblaze B2 Media Storage + Cloudflare CDN	**$0.00/mo** (~5 GB stored < 10 GB B2 free tier; 85% CDN cache hit rate; uncached Class B reads < 75k/mo free cap; $0 egress).	**$1.18/mo** (~160 GB accumulated storage = 150 GB billable \times \$0.006 = \$0.90 + 700k uncached reads \times \$0.40/\text{M} = \$0.28; 1.8 TB CDN egress = **$0.00**). *(Note: On Cloudflare R2 this would cost $2.25)*.	**$8.60/mo** (~1,100 GB accumulated storage \times \$0.006 = \$6.54 + 5.2M uncached reads = $2.06; 14 TB CDN egress = **$0.00**). *(Note: On R2 this would cost $16.35; on AWS S3 $1,285/mo)*.	**Dominant variable:** Cumulative media GB stored and CDN cache hit ratio. Guardrail: Cache-Control: immutable on cdn.swiftradish.app + 6h orphan purge.
4. Redis 7, Transactional Email & Observability	**$10.00/mo** (Managed Redis $10/mo; Resend Email free tier 3,000 emails/mo; Sentry Developer free tier).	**$25.00/mo** (Managed Redis $15/mo + Resend 10–20/mo for OTPs/statutory acks; Sentry free tier).	**$85.00/mo** (Redis HA $35/mo + Resend Pro $20/mo + Sentry Team $29/mo).	**Dominant variable:** Transactional email volume. Guardrail: Use email strictly for Auth OTPs, password resets,

Cost Driver / Infrastructure Layer	Low Scenario (Seed Beta: 1,000 MAU / 250 DAU, 1,500 posts/mo)	Base Scenario (Growing Community: 10,000 MAU / 3,000 DAU, 20,000 posts/mo)	High Scenario (Scaled Traction: 50,000 MAU / 15,000 DAU, 120,000 posts/mo)	Dominant Variable & Cost Guardrails
				and statutory ticket acks.
Total Monthly Infrastructure Run-Rate	**$24.00/mo** (Dev/Alpha) | **$49.00/mo** (Prod Launch with Supabase Pro)	**$116.68 / month** (~$0.012 per MAU)	**$368.85 / month** (~$0.0074 per MAU)	Hard billing alerts configured at **$75**, **$150**, and **$400/mo** across Supabase, B2, and compute hosts.

M. Cursor-Friendly Implementation Roadmap

13.1 Repository Hygiene and AI Coding Agent Rules (Grok / Composer 2.5)

Because the solo builder is operating with Cursor Pro using Grok and Composer 2.5 ([USER DECISION]), the implementation plan avoids sprawling multi-subsystem prompts that exceed working context windows. In Prerequisite Step 0, before generating application code, the builder must verify local repository status (git status), inspect .gitignore (ensuring .env, .env.local, .venv/, node_modules/, and .next/ are ignored), commit DESIGN.md, AGENTS.md, and .env.example (containing only placeholder keys, never real Supabase service-role keys or Backblaze B2 application keys), link the local repository to the created GitHub remote (git remote add origin https://github.com/<owner>/Swiftradish.git), and push the clean baseline (git push -u origin main). Every subsequent slice is executed on a dedicated feature branch (feat/slice-01-foundation, etc.), instructing Composer 2.5 or Grok to modify at most 3 to 5 files per prompt, run pytest, tsc --noEmit, and playwright after every sub-task, and commit immediately once tests pass—strictly prohibiting any claim that a test passed unless the terminal command was actually executed and exited 0.

13.2 Staged End-to-End Vertical Slice Roadmap

Slice ID & Name	User-Visible Outcome	Included vs. Excluded Scope	Prerequisites & High-Level Modules Involved	Data & API Contracts Delivered	Mandatory Executable Tests & Acceptance Criteria	Security, Observability & Git Checkpoint Gate
Slice 0: Cloud Verification, Repo Baseline &	Local and cloud environments boot cleanly; FastAPI	**Included:** Git remote setup, Docker Compose	User's GitHub repo, Supabase project URL/keys,	GET /api/v1/health returning {"status": "ok",	pytest tests/core/test_preflight.py verifying server	Verify zero secrets in git diff; tag checkpoint v0.1.0-slice0.

Slice ID & Name	User-Visible Outcome	Included vs. Excluded Scope	Prerequisites & High-Level Modules Involved	Data & API Contracts Delivered	Mandatory Executable Tests & Acceptance Criteria	Security, Observability & Git Checkpoint Gate
DB Preflight	/api/v1/health verifies Supabase Postgres, Redis, B2 connectivity, and Alembic HEAD match.	(redis, minio local mock), Alembic init (0001_extensions.py enabling pg_trgm), DESIGN.md, SCR-01 static shell, SCR-40 error shells. **Excluded:** Auth, posts, media.	and Backblaze B2 bucket credentials inspected in .env. Modules: backend/src/core/, frontend/src/app/.	"db_revision": "<head>", "redis": "ok"} in < 200ms.	refuses to start when alembic_version mismatches HEAD, plus alembic upgrade head && alembic downgrade base && alembic upgrade head.	**Stop Gate:** Halt if Supabase pg_trgm or B2 S3 credentials fail.
Slice 1: 18+ Auth, ES256 JWKS Verification & Onboarding Identity	Users can sign up (with 18+ age gate & DPDP consent), verify email OTP, log in, claim a unique @username, and manage active sessions.	**Included:** SCR-04, SCR-05, SCR-06, SCR-07, SCR-08, SCR-33 (sessions), users & user_profiles tables. **Excluded:** Avatar upload (Slice 3), social graph (Slice 2).	Slice 0 + Supabase Auth ES256 JWKS URL verified. Modules: backend/src/modules/identity/, frontend/src/app/(auth)/.	POST /api/v1/auth/register, POST /api/v1/auth/verify-otp, GET /api/v1/users/check-username, GET/PATCH /api/v1/users/me.	Integration tests verifying: (a) <18 DOB returns 422 with 0 DB rows written; (b) ES256 JWT verified locally in <1ms; (c) HS256 or wrong-iss tokens rejected 401; (d) Playwright E2E signup -> handle claim.	Verify HttpOnly; Secure; SameSite=Lax cookies and PII-scrubbed auth logs. Merge feat/slice-01-auth -> tag v0.2.0-slice1.
Slice 2: Profiles, Hybrid Social Graph, Private Accounts &	Users can view/edit profiles (SCR-23/24), follow public accounts, request to	**Included:** SCR-23, SCR-24, SCR-2[span_122](start_span)[span_122](end_span)	Slice 1. Modules: backend/src/modules/social_graph/, frontend/src/app/(main)/u/	GET /api/v1/users/{username}/profile, POST/DELETE /api/v1/relatio	Integration tests verifying: (a) reciprocal ACCEPTED follow sets is_mutual =	Verify Two-Plane BOLA checks on private profiles. Merge

Slice ID & Name	User-Visible Outcome	Included vs. Excluded Scope	Prerequisites & High-Level Modules Involved	Data & API Contracts Delivered	Mandatory Executable Tests & Acceptance Criteria	Security, Observability & Git Checkpoint Gate
Atomic Blocking	follow private accounts (SCR-25), auto-form Mutual edges, and block/mute users (SCR-34).	5, SCR-34, user_relationships, user_blocks, user_mutes. **Excluded:** Posts and feeds.	[username]/.	ns/{userId}/follow, POST /api/v1/relationships/requests/{id}/decide, POST/DELETE /api/v1/safety/blocks.	TRUE on both edges; (b) private profile hides relations from non-followers (404); (c) blocking atomically deletes follow edges in both directions.	feat/slice-02-graph -> tag v0.3.0-slice2.
Slice 3: Direct-to-B2 Media Pipeline, pyvips Verification & Composer State Machine	Users can select 1–10 images, crop/reorder/add alt-text (SCR-16), upload directly to Backblaze B2 via presigned PUT (SCR-15), verify/strip EXIF on server, and inspect via Lightbox (SCR-14).	**Included:** SCR-14, SCR-15 (media tray), SCR-16, media_assets table, Cloudflare B2 Worker config, arq 6-hour orphan cleanup job. **Excluded:** Video uploads (deferred).	Slice 2 + Backblaze B2 CORS & Application Key configured. Modules: backend/src/modules/media/, backend/src/workers/media_tasks.py.	POST /api/v1/media/upload-intents, POST /api/v1/media/{id}/complete, DELETE /api/v1/media/{id}.	Playwright E2E test: select 3 images -> upload progress -> READY -> cancel 1 image -> verify EXIF GPS stripped -> verify SVG/polyglot & 100MP bomb rejected 422 -> verify orphan cron deletes unattached B2 objects.	Verify B2 bucket is private (allPrivate) and cdn.swiftradish.app returns Cache-Control: immutable. Tag v0.4.0-slice3.
Slice 4: Standard Posts, Co-Plot Coordination Cards, RSVPs,	Users can publish posts and Co-Plots (SCR-[span_156](start_span)[span_156](end_span)	**Included:** SCR-02, SCR-13, SCR-15, SCR-17, posts, co_plots,	Slice 3. Modules: backend/src/modules/posts/, co_plots/, messaging/.	POST/GET/DELETE /api/v1/posts, POST /api/v1/plots/{id}/rsvp, POST	Playwright E2E test: Host plants Co-Plot -> composer resets cleanly ->	Verify Idempotency-Key on post creation, FOR UPDATE lock on

Slice ID & Name	User-Visible Outcome	Included vs. Excluded Scope	Prerequisites & High-Level Modules Involved	Data & API Contracts Delivered	Mandatory Executable Tests & Acceptance Criteria	Security, Observability & Git Checkpoint Gate
Shared Carousels & Plot Thread	15), view public/authenticated detail (SCR-02, SCR-13), RSVP (Going/Maybe), submit collaborative carousel slides, approve slides (SCR-17), and chat in the Co-Plot thread.	co_plot_participants, conversations & messages (scoped to Co-Plot thread with WebSocket + REST fallback). **Excluded:** Standalone 1:1 DM inbox (Stage 2).		/api/v1/plots/{id}/submissions, POST /api/v1/plots/{id}/submissions/{s[span_157](start_span)[span_157](end_span)ubId}/decide, WS /ws.	Immediate 2nd post upload succeeds -> Mutual RSVPs Going & submits photo -> Host approves in SCR-17 -> Slide appears in carousel -> Plot thread deduplicates reconnect messages.	RSVP capacity, and client_msg_id deduplication. Tag v0.5.0-slice4.
Slice 5: Radish Patches, Dual-Mode Feeds (My Patch & Fresh Harvest), Reactions, Comments & Hybrid Search	Users can complete Patch onboarding (SCR-09), browse chronological (SCR-11) and explainable discovery (SCR-12) feeds, join Patches (SCR-03, SCR-21), react, write 3-level comments, bookmark, and search (SCR-19, SCR-20).	**Included:** SCR-03, SCR-09, SCR-11, SCR-12, SCR-19, SCR-20, SCR-21, patches, patch_members, post_reactions, comments, bookmarks. **Excluded:** 3-channel Patch live chat lounge (Stage 2).	Slice 4. Modules: backend/src/modules/feed/, patches/, search/.	GET /api/v1/feed/following, GET /api/v1/feed/discover, POST/GET /api/v1/patches/{slug}, POST /api/v1/posts/{id}/reactions, POST /api/v1/posts/{id}/comments, GET /api/v1/search.	Integration & Playwright tests: (a) keyset pagination produces 0 duplicate/skipped posts when new posts arrive; (b) SQL query count \le 3 (0 N+1 queries); (c) comment depth >2 rejected; (d) pg_trgm typo search "urbn sktching" matches "Urban Sketching".	Verify blocked/muted users never appear in feeds or search results. Merge feat/slice-05-feeds-search -> tag v0.6.0-slice5.

Slice ID & Name	User-Visible Outcome	Included vs. Excluded Scope	Prerequisites & High-Level Modules Involved	Data & API Contracts Delivered	Mandatory Executable Tests & Acceptance Criteria	Security, Observability & Git Checkpoint Gate
Slice 6: Activity Notifications, Statutory Grievance Flow (IT Rules 2021), Admin Triage & Account Deletion	**Completes MVP Launch Core:** Users receive deduplicated in-app notifications (SCR-31), file statutory safety reports (SCR-36), track tickets/appeals (SCR-38), export/delete accounts (SCR-37), and admins triage reports (SCR-39).	**Included:** SCR-31, SCR-36, SCR-37, SCR-38, SCR-39, /legal/grievance, notifications, safety_reports, audit_events. **Excluded:** Stage 2 enhancements.	Slice 5. Modules: backend/src/modules/notifications/, safety/.	GET/POST /api/v1/notifications, POST /api/v1/safety/reports, GET /api/v1/safety/my-reports, POST /api/v1/safety/appeals, POST /api/v1/admin/moderation/cases/{id}/resolve, POST /api/v1/account/schedule-deletion.	E2E verification: (a) reaction rollup deduplicates via dedupe_key; (b) submitting a report hides content immediately and generates ticket ack in <60s; (c) UPDATE/DELETE on audit_events fails via DB trigger; (d) account deletion hides profile immediately and schedules B2 purge.	Run full 5-gate release audit (pytest, BOLA matrix, Playwright desktop/mobile + axe-core, k6 load test). **Tag v1.0.0-mvp-launch**.
Stage 2 Slices (Slices 7–9 — Post-MVP Launch Fast-Follow)	**Slice 7:** Standalone 1:1 & Group DM Inbox + Message Requests Quarantine (SCR-27–30). **Slice 8:** Co-Plot Calendar .ics Hub (SCR-18), Bookmark	Built incrementally after v1.0.0-mvp-launch is live with initial beta cohorts.	v1.0.0-mvp-launch telemetry and user feedback.	Preserves all Stage 2 schemas and endpoints defined in Sections D, H, and I.	Each Stage 2 slice requires passing the same 5 release gates before merge.	Feature-flagged per slice (dm_inbox_enabled, web_push_enabled).

Slice ID & Name	User-Visible Outcome	Included vs. Excluded Scope	Prerequisites & High-Level Modules Involved	Data & API Contracts Delivered	Mandatory Executable Tests & Acceptance Criteria	Security, Observability & Git Checkpoint Gate
	Folders (SCR-26), Patch Steward Console (SCR-22), PWA Install/Push Primer (SCR-10, SCR-35). **Slice 9:** Badge Engine (SCR-32) & Post-Beta Video/Sprouts Evaluation.					

N. Decision Register and Unresolved Questions

All consequential product, design, architectural, and legal choices are logged below with explicit status distinguishers ([USER DECISION], [RECOMMENDATION], or User Decision / Sign-Off Required).

Decision ID & Topic	Status & Label	Reconciled Choice	Rationale & Evidence	Dependencies & Revisit Trigger
DEC-01: MVP Launch Surface Boundary (22 Core Screens vs 40 Total)	**[RECOMMENDATION] — User Sign-Off Required** (Confidence: High)	Launch MVP Core across **Slices 1–6 (22 screens)** featuring unified Co-Plots (RSVP + Shared Carousel + Plot Thread), Patches, Dual Feeds, and Statutory Safety; stage standalone 1:1 DM inbox (SCR-27–30), Web Push (SCR-10), and badge engine (SCR-32) into	Protects solo builder using Cursor Pro (Grok / Composer 2.5) from multi-subsystem failure while delivering 100% of the Co-Plot thesis.	Unlocks Stage 2 immediately after Slice 6 passes v1.0.0-mvp-launch release gates.

Decision ID & Topic	Status & Label	Reconciled Choice	Rationale & Evidence	Dependencies & Revisit Trigger
		Stage 2 (Slices 7–9).		
DEC-02: Primary Media Object Storage (Backblaze B2 vs Cloudflare R2)	**[USER DECISION] + [RECOMMENDATION] — Accept & Configure** (Confidence: High)	Retain the user's **existing Backblaze B2 bucket** ($0.006/GB/mo, presigned PUT) fronted by **Cloudflare CDN + Signing Worker (Bandwidth Alliance)** for $0.00 egress; keep R2 as a zero-code-change S3 fallback.	Honors user's provisioned B2 bucket, cuts storage cost by 60% vs R2 ($0.006 vs $0.015/GB), and eliminates egress fees via Bandwidth Alliance.	Requires completing the 5-step B2 + Cloudflare Worker setup in Slice 0/3; revisit only if B2 regional latency from India exceeds 200ms on uncached misses.
DEC-03: Primary Database & Auth Stack (Supabase Postgres + Local ES256 JWKS)	**[USER DECISION] + [VERIFIED] — Accept** (Confidence: High)	Use the user's **existing Supabase project** for PostgreSQL 16 (via PgBouncer port 6543 + Alembic) and Supabase Auth (ES256 JWKS verified locally in FastAPI via pyjwt[crypto]).	Eliminates separate database vendor costs at launch and verifies tokens in <0.5ms while blocking algorithm confusion (CVE-2026-10226 6).	Upgrade Supabase from Free $(0) to Pro($ 25/mo) before inviting external beta users so the DB never pauses after 7 days of inactivity.
DEC-04: Visual Identity & Mascot Scope	**[RECOMMENDATION] — Accept Simplified MVP** (Confidence: High)	Adopt **"Nocturnal Botanical Studio"** (#0B0D11 dark canvas + Daylight Parchment light mode) and ship **8 static layered Pip SVGs** in MVP Core, deferring 30/216 poses and Lottie animations to Stage 2.	Delivers warm, distinctive branding and gallery-grade photo contrast without blocking core engineering on 216 custom illustrations.	Revisit full mascot expansion only after D30 cohort retention exceeds 18%.
DEC-05: 18+ Age Gate & DPDP Rules 2025 Compliance	**[RECOMMENDATION] — User & Legal Sign-Off Required**	Enforce strict 18+ age gate at /register, standalone	Aligns with MeitY's Nov 13–14, 2025 notification of the DPDP Rules 2025	Consult qualified Indian legal counsel prior to commercial

Decision ID & Topic	Status & Label	Reconciled Choice	Rationale & Evidence	Dependencies & Revisit Trigger
Posture	(Confidence: High)	itemized DPDP consent notice, 72-hour breach-notification runbook (Rule 7), and IT Rules 2021 24h/72h/15d grievance SLA queues.	and amended IT Rules 2021 Rule 3.	scaling on DigiLocker age-token verification and Eighth Schedule language translations.

Unresolved Questions Requiring User Confirmation Before Coding Slice 1

Before initiating Slice 1 implementation, three unresolved questions require explicit user confirmation. First, regarding **Cloud Region Inspection ([UNVERIFIED])**, the user must confirm which exact regions were selected when creating the **Supabase project** (ap-south-1 Mumbai versus US or EU) and the **Backblaze B2 bucket** (us-west, us-east, or eu-central), as well as whether a custom domain on Cloudflare is already active for the Bandwidth Alliance Worker. Second, regarding the **Named Grievance Officer ([UNVERIFIED])**, the user must provide the name, designation, email address, and physical contact address that will populate the statutory /legal/grievance disclosure required under India's IT Rules 2021 Rule 3(2). Third, regarding **MVP Slice Boundary Approval ([USER DECISION] Required)**, the user must confirm approval of the staged **22-screen MVP Launch Core (Slices 1–6)** with Co-Plot threads included in Slice 4 and the standalone 1:1 DM inbox (SCR-27–30) staged in Slice 7.

O. Phase 3 Independent Audit Handoff Pack

To enable the Phase 3 independent auditor to stress-test every assumption, claim, and contract in this specification without blind spots, the complete audit handoff dossier is synthesized below.

15.1 Source Inventory and Reconciled Artifacts

The primary input artifacts governing this specification are SWIFTRADISH_PHASE_2_GEMINI_RESEARCH_PROMPT.md (canonical constraints and Phase 2/3 prompt rules) and Swiftradish Product and Engineering Blueprint.pdf (the prior 73-page research artifact). These internal artifacts have been reconciled against external primary verification sources covering Backblaze B2 S3 compatibility, presigned PUT versus POST constraints, and Cloudflare Bandwidth Alliance Worker documentation; Cloudflare R2 official pricing; Supabase ES256 JWKS documentation and PyJWKClient security advisories (CVE-2026-102266); India's DPDP Act 2023 and DPDP Rules 2025 (notified Nov 13–14, 2025 via G.S.R. 843(E), 844(E), 846(E)); and India's IT Rules 2021 (as amended Oct 28, 2022 via G.S.R. 794(E) and Apr 6, 2023 via G.S.R. 275(E)).

15.2 High-Impact Claims and Architectural Choices the Phase 3 Auditor Must Challenge

The Phase 3 independent auditor must directly challenge five high-impact architectural and operational determinations. First, under **Backblaze B2 Latency for Indian Users (REC-02 / DEC-02)**, Backblaze B2 operates data centers in US-West, US-East, and EU-Central with no Mumbai (ap-south-1) origin. While Cloudflare CDN caches 85%+ of public image reads at Indian edge PoPs (cdn.swiftradish.app) after the initial fetch, direct browser presigned PUT uploads and uncached first reads traverse from India to the US or EU. The auditor must evaluate whether cross-continent presigned PUT upload latency (~180ms–250ms RTT) is acceptable for 1–5MB images in Slice 3 or whether R2's automatic APAC placement warrants switching the S3 endpoint if upload p95 exceeds 2.5s.
Second, under **Supabase PgBouncer Transaction Pooling vs. SQLAlchemy Async (REC-05 / DEC-03)**, Supabase's transaction pooler (port 6543) rejects prepared statements unless asyncpg is explicitly configured with connect_args={"statement_cache_size": 0}, and session-level advisory locks or LISTEN/NOTIFY cannot run over transaction-mode PgBouncer. The auditor must verify that every SQL query in Section H—including SELECT ... FOR UPDATE inside explicit async with session.begin(): transactions—is 100% compatible with PgBouncer transaction mode.
Third, under **Co-Plot Thread vs. Full DM Inbox Staging (REC-01 / DEC-01)**, Sections C and M stage the **Co-Plot Contextual Thread** in MVP Core (Slice 4) while deferring the standalone **1:1 Direct Message Inbox & Message Requests Quarantine (SCR-27–30)** to Stage 2 (Slice 7). The auditor must challenge whether early beta cohorts can coordinate effectively using only Co-Plot threads and public/Patch comments, or whether a minimal 1:1 mutual-only DM drawer is required prior to public beta.
Fourth, under **Synchronous pyvips Verification vs. Worker Queue Offload (Section 8.2 & 9.1)**, processing up to three WebP variants synchronously inside POST /api/v1/media/{id}/complete via asyncio.to_thread() simplifies the client state machine (VERIFYING to READY in a single request) but bounds concurrent image processing to the FastAPI container's CPU and RAM. The auditor must stress-test the exact concurrency threshold (such as asyncio.Semaphore(4) per worker process) at which complete must spill over to 202 Accepted plus arq background processing.
Fifth, under **Legal & Statutory Completeness (Section 10.2)**, the auditor must verify that the database schema (users.consent_version, safety_reports.sla_tier, audit_events append-only trigger, and 180-day B2 quarantine retention) leaves zero statutory gaps under India's IT Rules 2021 Rule 3 and the notified DPDP Rules 2025 ahead of the May 14, 2027 compliance deadline.

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